

Video Astronomie mit Smart Telescopen?

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Heppenheim

22. September 2026



Seestar S50

Beobachtung der Sternbedeckung von (5368) Vitagliano am 12.03.2026 mit Bernd Gährken nördlich Regensburg

- Video mit Seestar app
- Zeitstempel?? 1pps Blitzer
- Filter Occult Watcher
- Signal-Rausch-Verhältnis?
- Magnituden des S50 bei Video
- Ereignis Filterung mit Excel
- DAMIT / ISAM
- Expedition
- Ergebnisse

International Occultation Timing Association European Section

2026 May 04 ~20:17 UT: (136108) Haumea occults UCAC4 524-056397 (14.7 mag)

A very important occultation, a TNO with rings and two moons!!!

<<< Have a look here >>>

Jupiter Trojan (617) Patroclus / Menoetius occults an 11.9 Vmag star across Europe on 2026 Jan 4st

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Phemu Events for Saturn, 2025-2026

Only a few events for the period 2025 to 2026 are observable with a reasonable chance of success. More infos you can find here.

Occultation by the Jovian satellite Ganymede in the early morning of the 14th of October 2025 - A European campaign

More infos and a meeting you can find here.

Observation report

Successful Observation of TNO 2013 LU28 Stellar Occultation in the far

Newest JOA ready for download!



Report for the Seestar telescope written by Christian Weber

This article describes aspects of the use of the

At the moment we don't support "**X**" anymore.

Here you can find us on **Mastodon**

The XLV ESOP meeting will be held in Toulouse (France), from 22 till 23 August 2026, with extra days for excursions. All the occultation community is cordially invited.

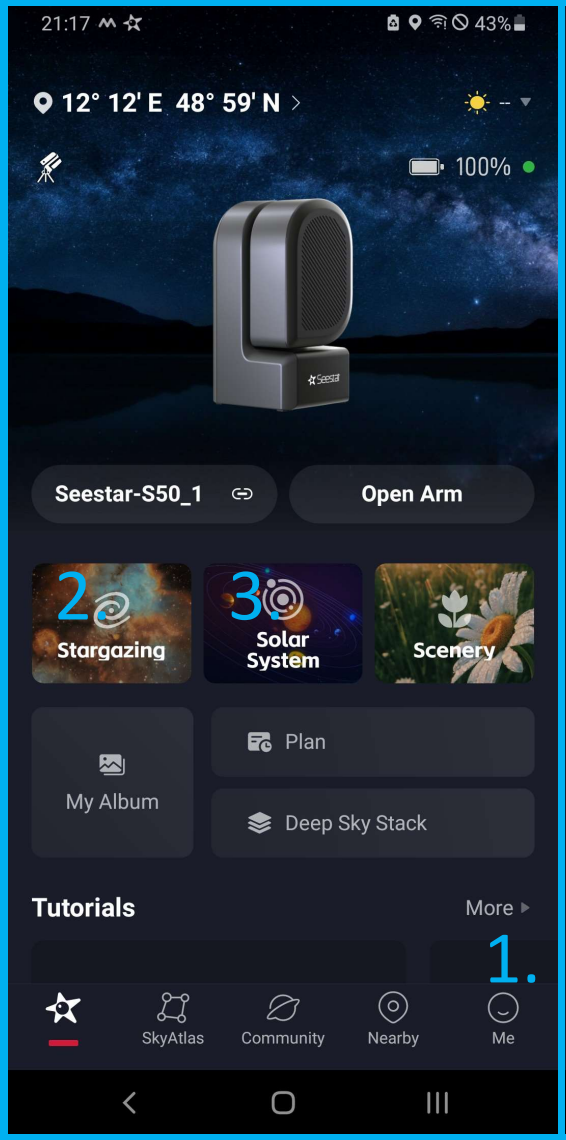
The Journal for Occultation Astronomy

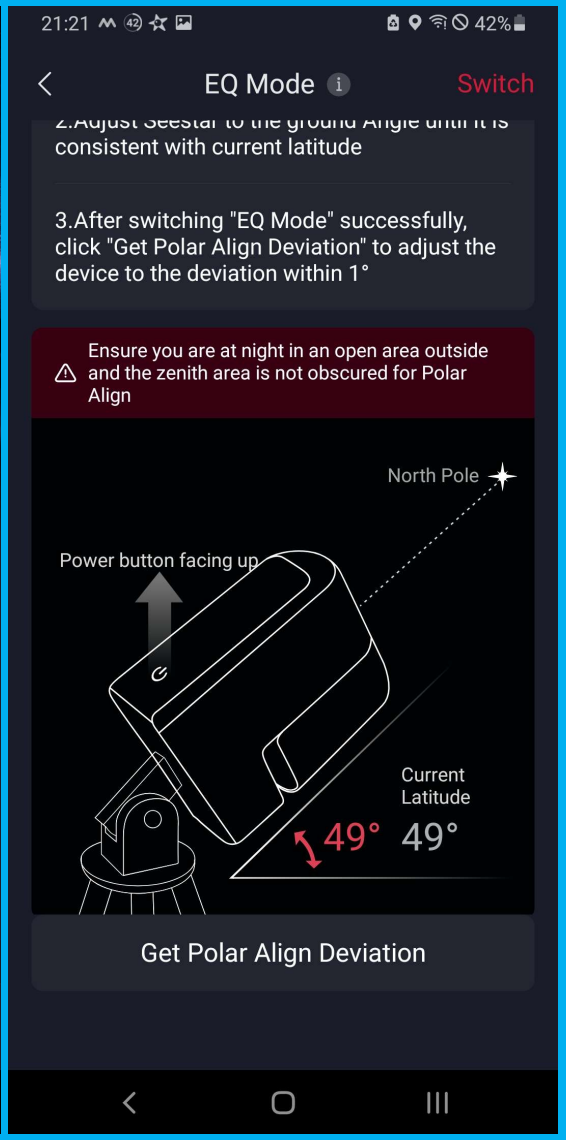
is now free for all!

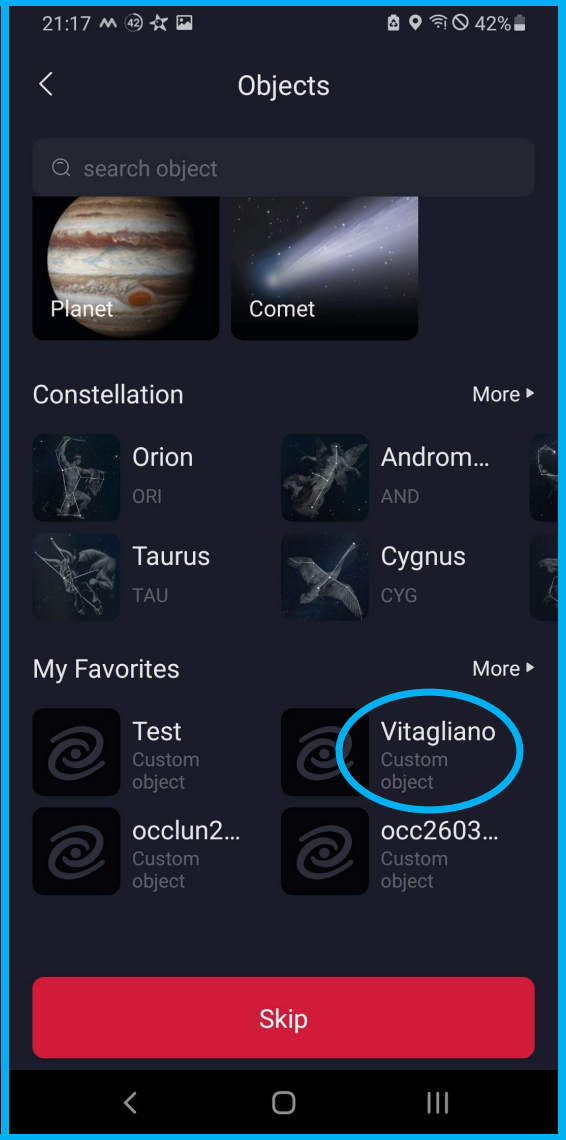
Call for Observations
Occultations of Stars by Minor Planets

Call for Observations

ursus
Internetzugriff

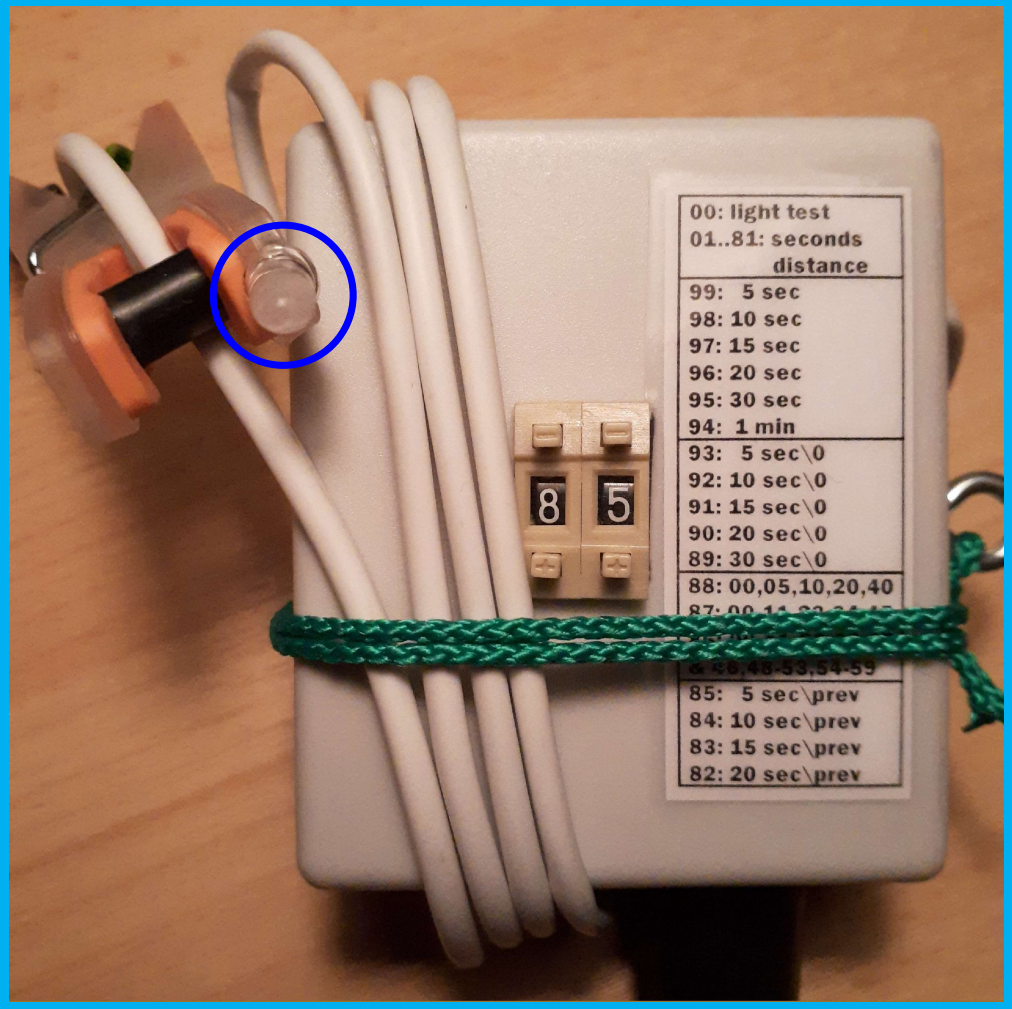

21:17
43%
12° 12' E 48° 59' N
100%
Seestar-S50_1
Open Arm
2
Stargazing
3
Solar System
Scenery
My Album
Plan
Deep Sky Stack
Tutorials
More
1.
SkyAtlas
Community
Nearby
Me


21:21
42%
EQ Mode
Switch
consistent with current latitude
3.After switching "EQ Mode" successfully, click "Get Polar Align Deviation" to adjust the device to the deviation within 1°
Ensure you are at night in an open area outside and the zenith area is not obscured for Polar Align
North Pole
Power button facing up
Current Latitude
49°
49°
Get Polar Align Deviation


21:17
42%
Objects
search object
Planet
Comet
Constellation
More
Orion
ORI
Androm...
AND
Taurus
TAU
Cygnus
CYG
My Favorites
More
Test
Custom object
Vitagliano
Custom object
occlun2...
Custom object
occ2603...
Custom object
Skip


12:16
96%
Neptune
98%
Slow
Fast
Track
4. Video und RAW!
=> AVI!
Raw
1x
2x
4x
Photo
Video

Timestamp? ->> 1 peak per second flash unit ->> task solved!



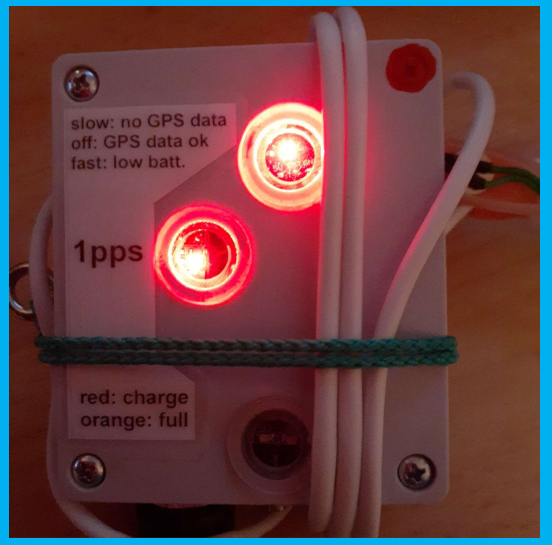
optisch! – unter 0,1ms Verzögerung!

an jedem Videosystem einsetzbar!

40ms Blitz, viele Blitzmuster

5 Stunden Betrieb

#85: 40ms Blitz alle 5 Sekunden \ 55te



Auf GITHUB:

- Bauanleitung
- Bedienungsanleitung
- firmware code (Arduino nano)
- hardware PCB (EAGLE)

DIY! ->> https://github.com/leontodon1/1pps_flasher

Zeitstempel am S50

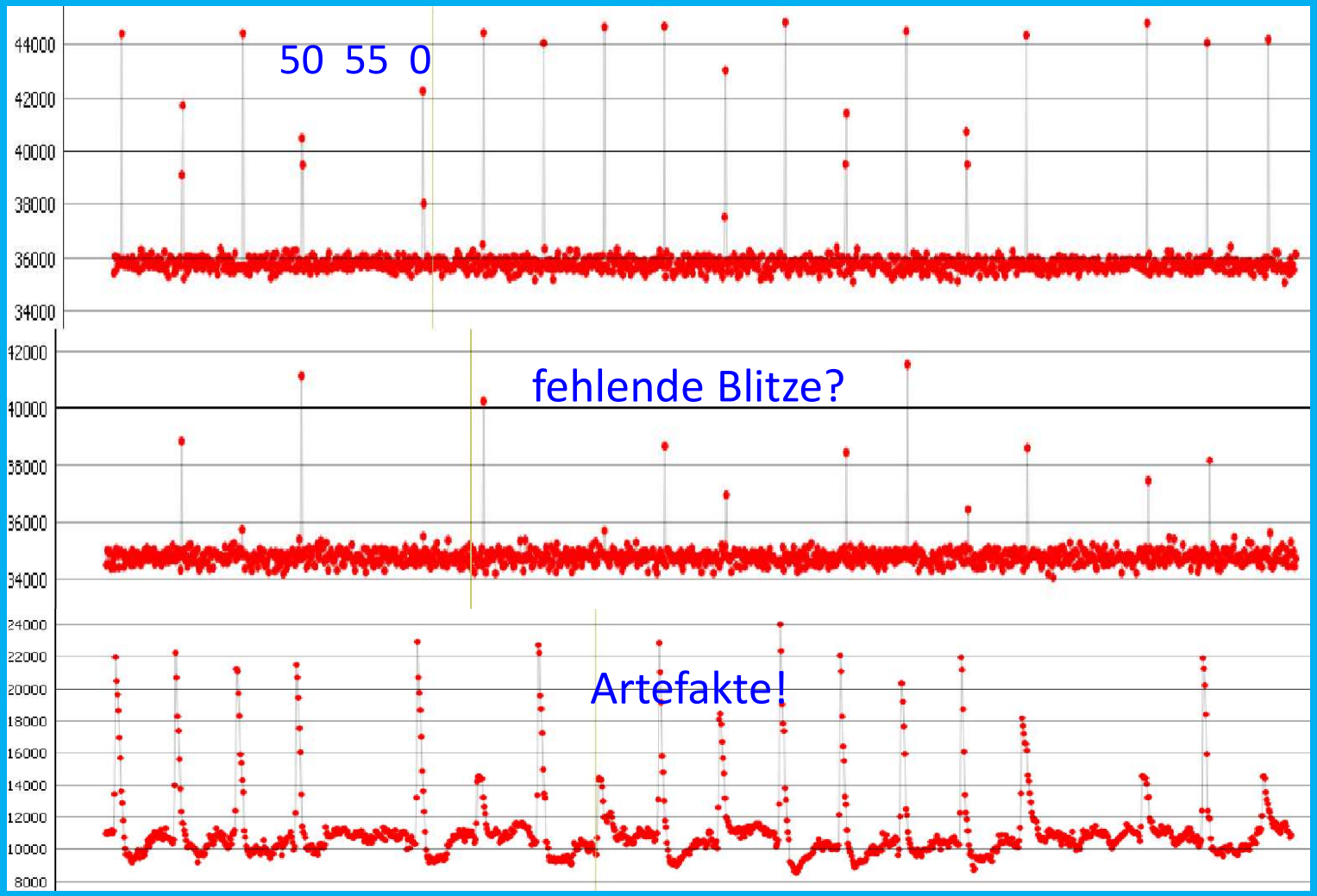
avi vs. mp4 (1x, #85)

avi „RAW“ 80ms

avi 35ms (1x)

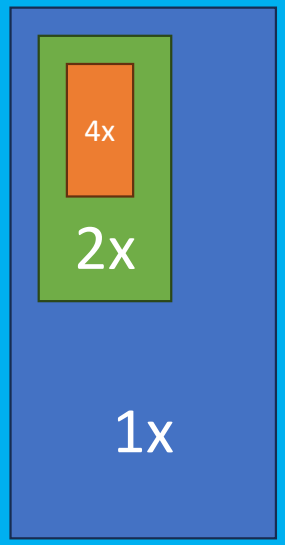
mp4 35ms

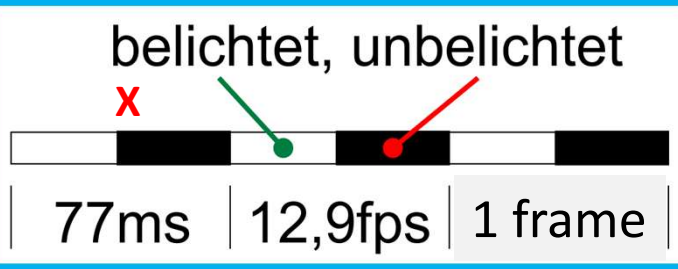
PyMovie / PyOTE

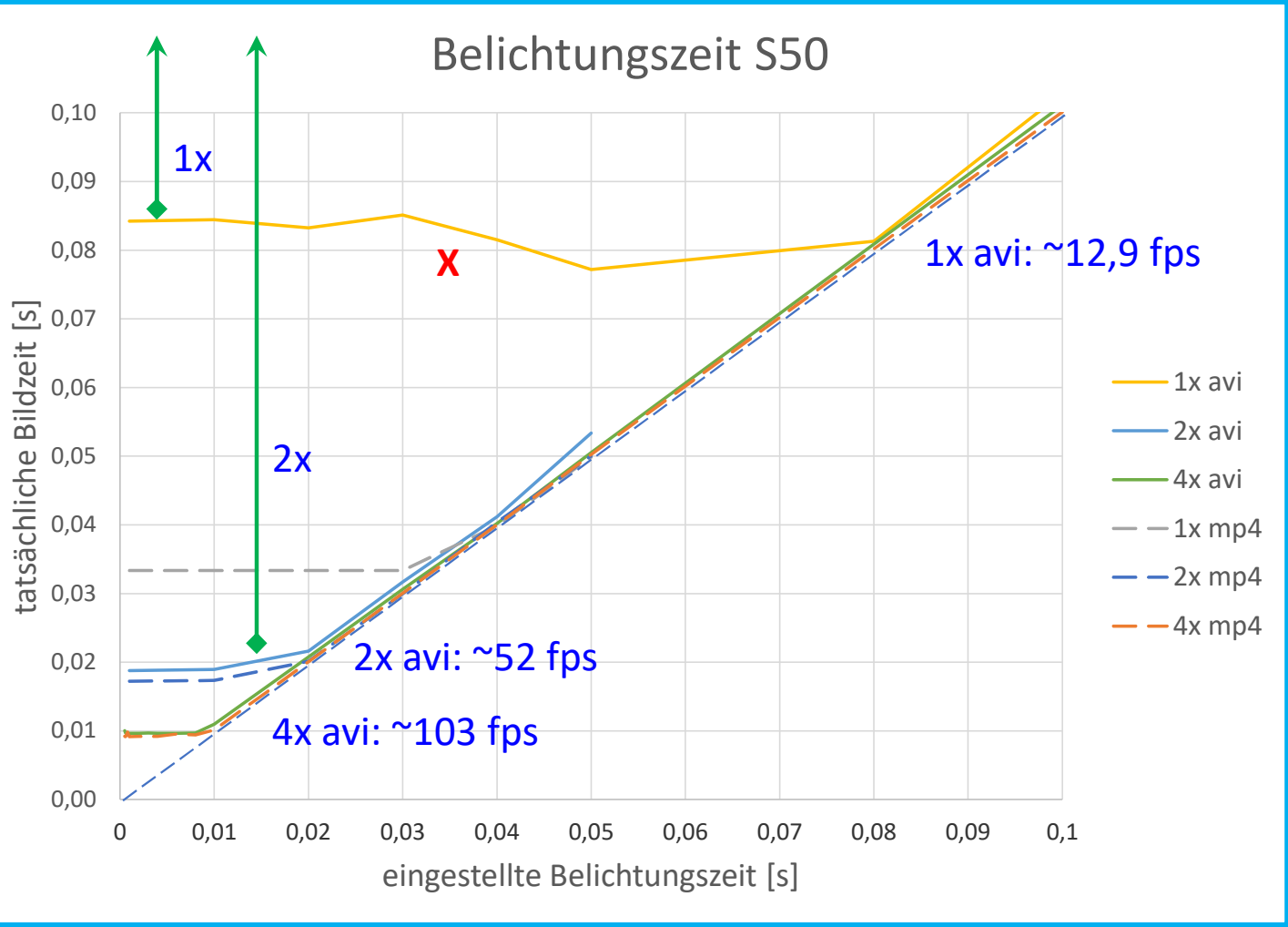


Achtung Falle! Belichtungszeit und fps beim S50

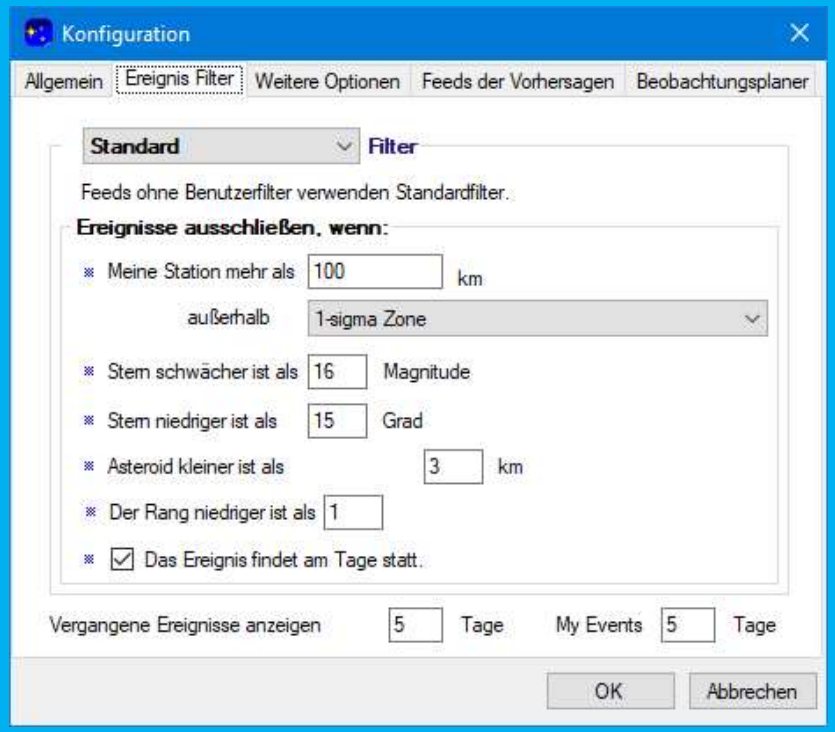
FOV Formate







Occult Watcher - Ereignis Filter



Benötigte Filter für S50 mobil:

Distanz Erdoberfläche < 30 km
drop > 2,5mag

gesuchtes Ergebnis:

- Bilder maximal möglich (frames_max ≥ 3)
  => Filter!
- kleinst mögliche Belichtungszeit t_exp_min (SNR > 3)
  Martin Gutekunst, 2025 ESOP 44 in Posen/PL

Parameter für t_exp_min, frames_max

Optik: Brennweite, Öffnung
Kamera: Pixelgröße, binning, Empfindlichkeit
OW: Dauer, Helligkeit, mag drop, Distanz

max. Magnituden beim S50 - h~45°

Grenzgröße

Nachweisgrenze :

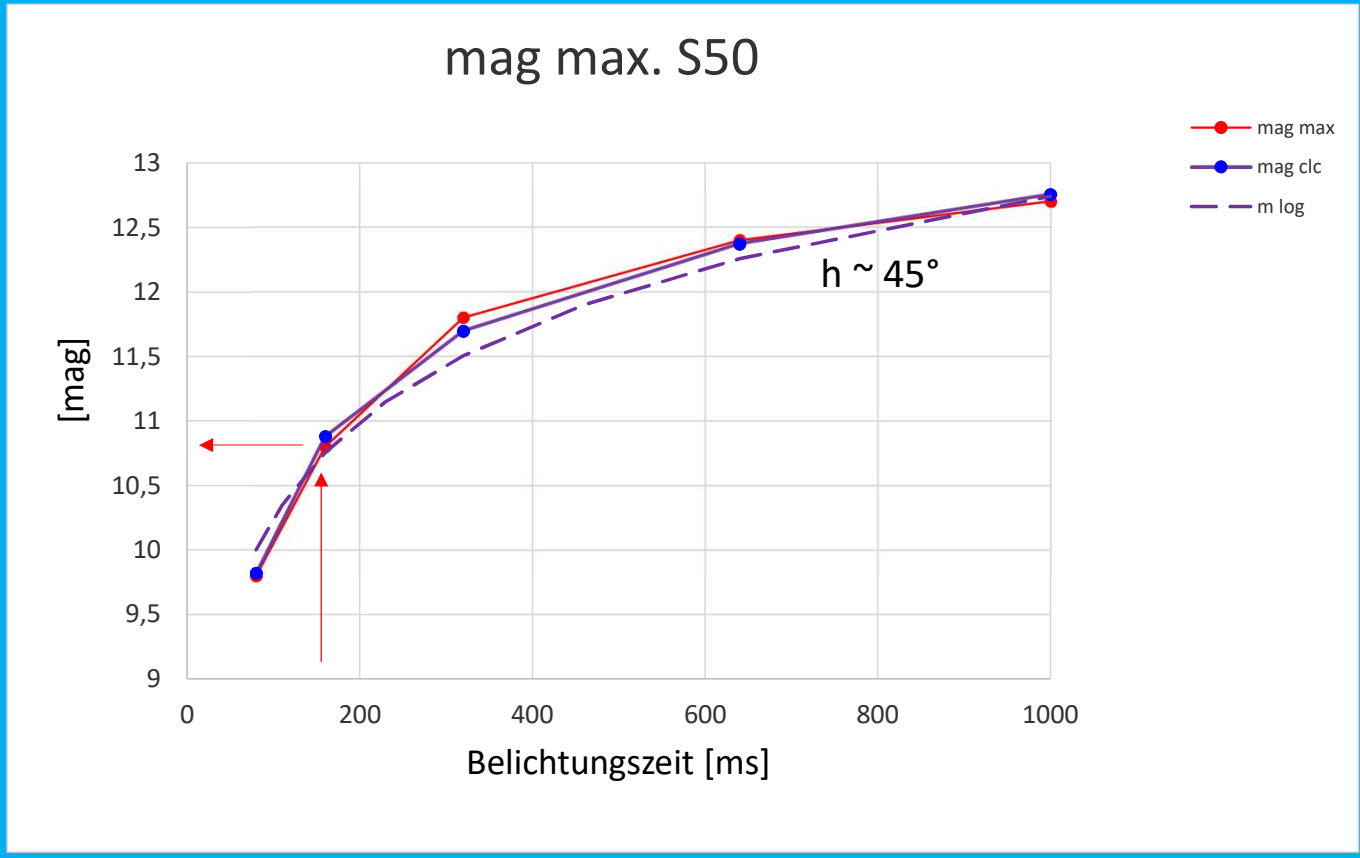
SNR << 3

experimentell ermittelt

$$m_{log} = 10 + \frac{ln\left({}^{t_{\exp(ms)}}/_{80}\right)}{ln(100^{\,0,2})}$$

empirische Formel(h ~ 45°):

.. nur anhand Belichtungszeit



$$m_{\max} = (-20500 + 6800 * ln(t_{\exp(ms)}))^{0,25}$$

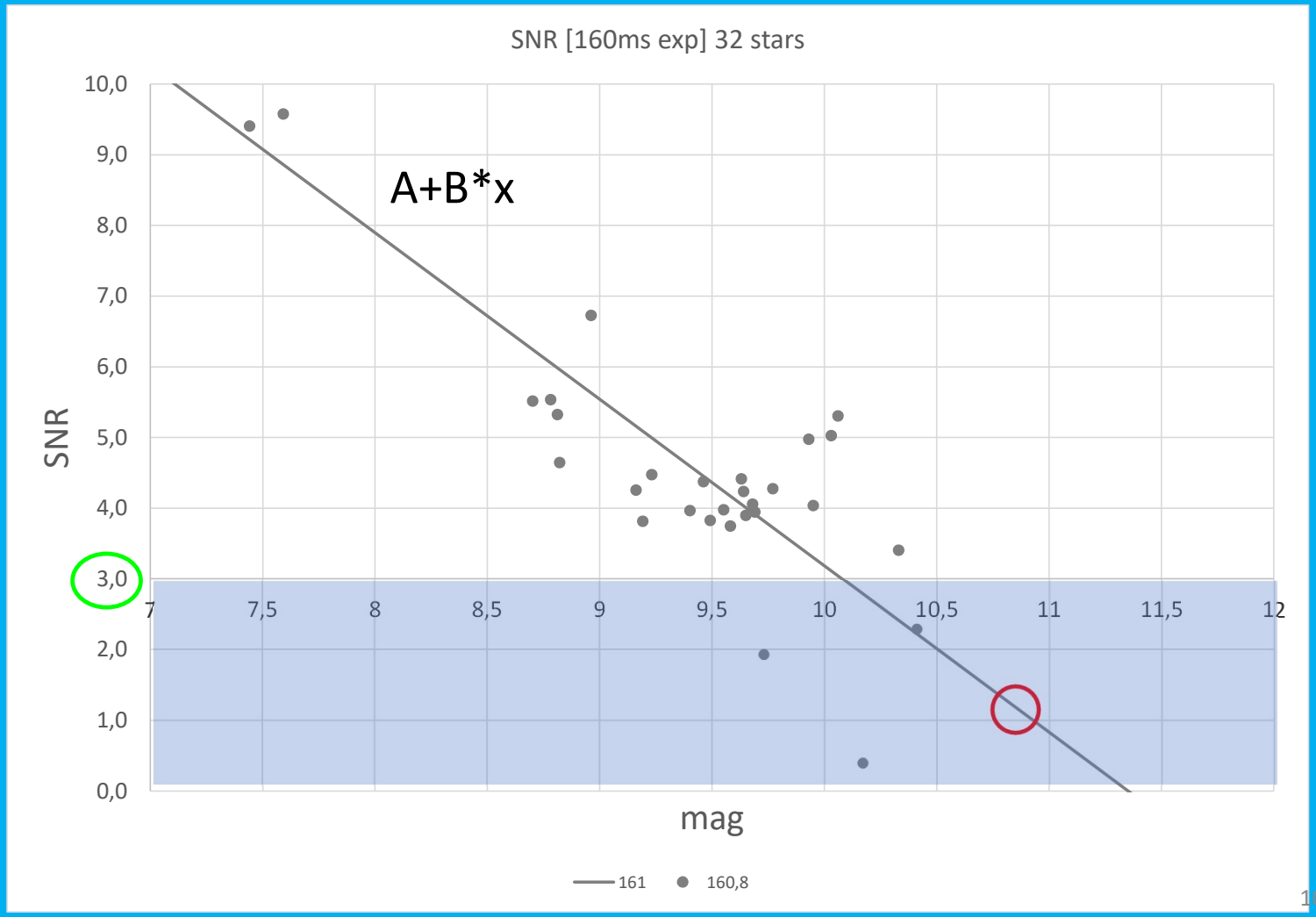
$form$: $(A + B * ln(t))^{0,25}$ empirisch

Signal-Rausch-Verhältnis S50

160ms Video

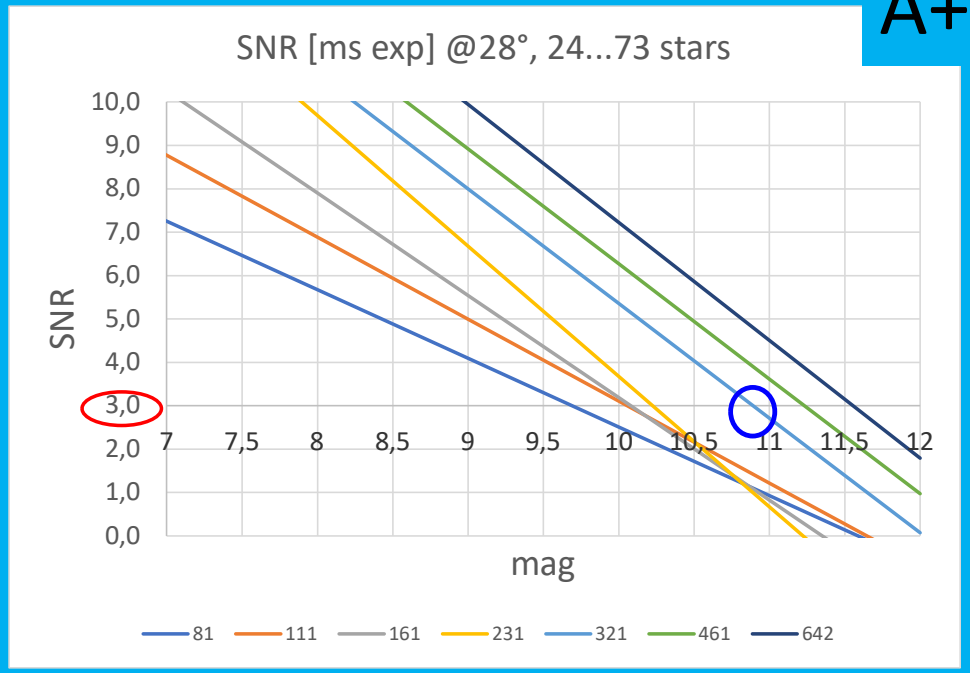
ca. 400 Bilder

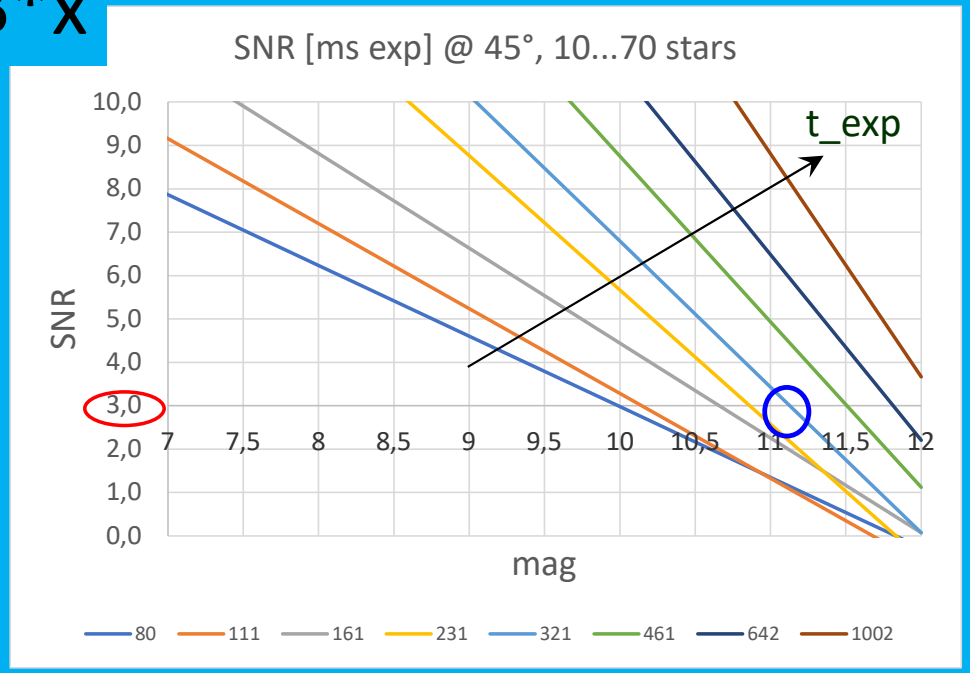
Auswertung: Lichtkurven von 32 Sternen



Signal-Rausch-Verhältnis S50

A+B*x





Extinktion 28° versus 45°: ca. 0.3 mag

Extinktion 45°: ~ 0,1 mag

Magnituden beim S50 @ SNR=3 & Extinktion

$m_{Extinktion} = k * A * h^B$

$k = 0{,}2\ vis.\ (0{,}15\ cam.)$

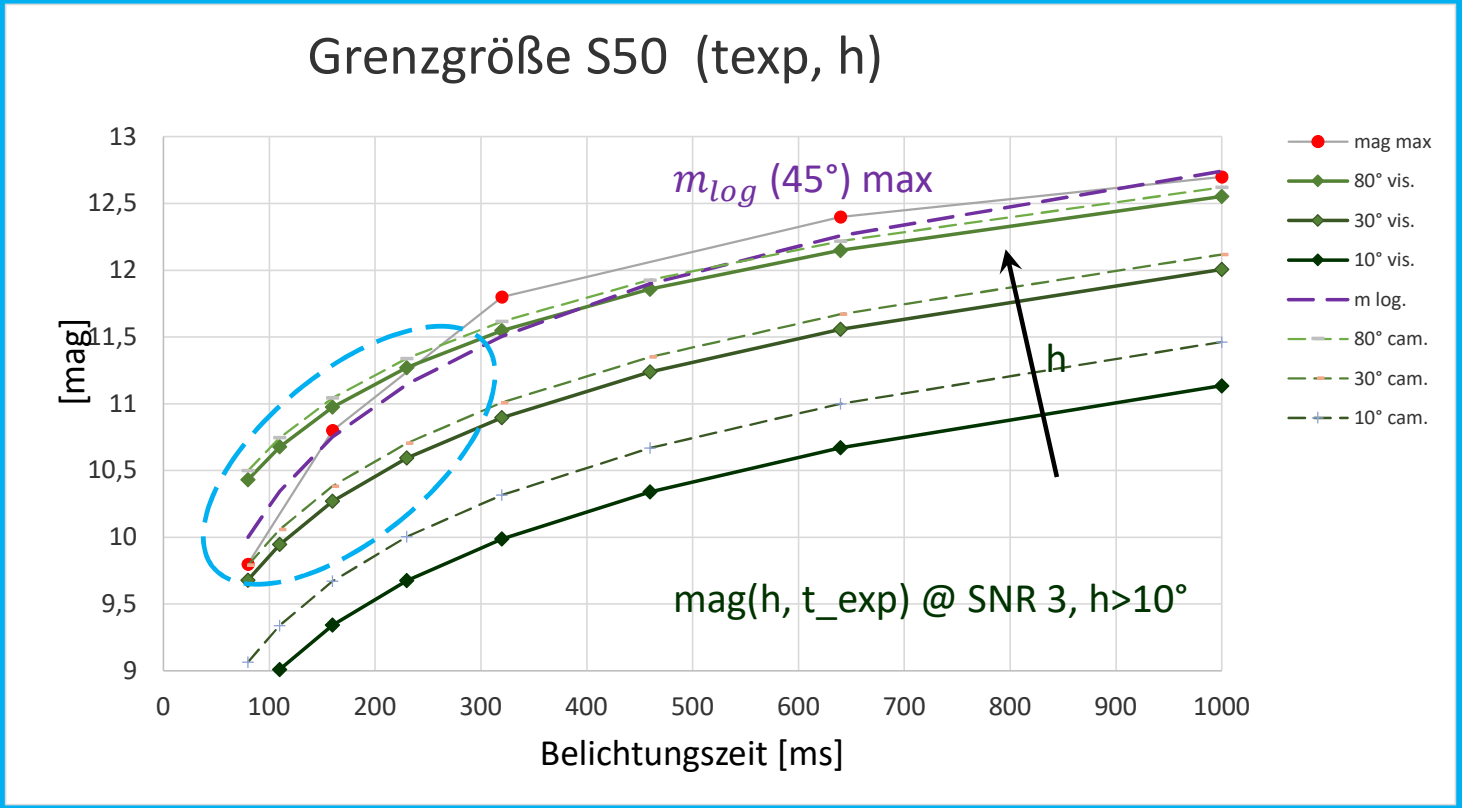
$A = 38{,}6 \quad B = -0{,}85$

$m_{log}(t_exp)$

Auswertbarkeit nur größer

SNR = 3 !

Belichtungszeit, Sternhöhe



h > 10°: $mag(t_{exp}, h) = A(h) * t_{exp}^{B(h)} - Extinktion$

$mag(S50, @SNR = 3) = (6{,}6 + 0{,}014 * h) * t_{\exp(ms)}^{(0{,}088-0{,}0002*h)} - m_{Extinktion}$

$mag(S50, @SNR = 3) = (7{,}9 - 0{,}014 * (90 - h)) * t_{\exp(ms)}^{(0{,}07+0{,}0002*(90-h))} - (0{,}2 + 0{,}0035 * e^{0{,}07*(90-h)})$

Auszug Occult Watcher - 3 Wintermonate: 2800 Ereignisse

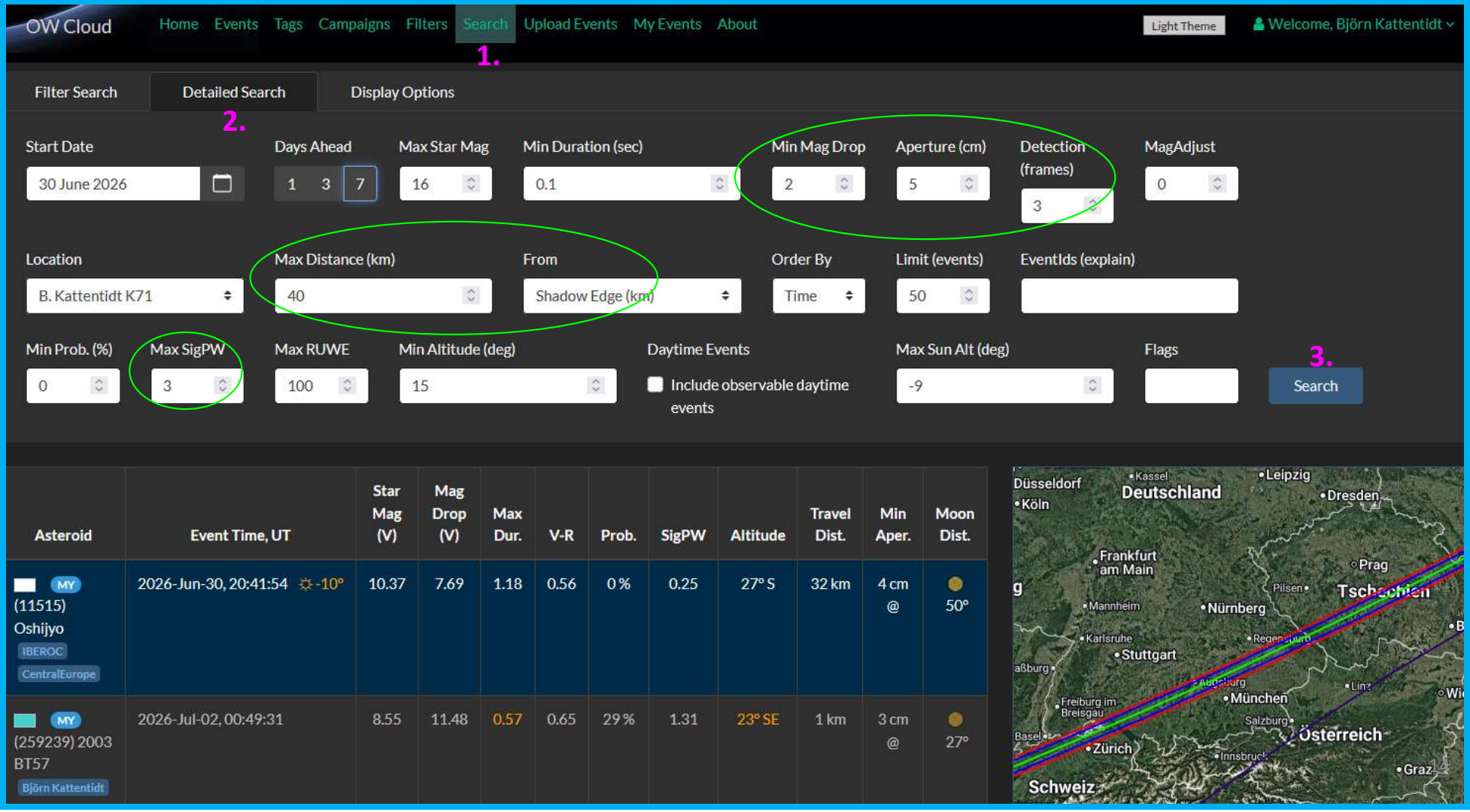
< 10mag, 5km Größe, 30km Distanz : 23 Ereignisse (angezeigt)

< 11mag, 40km Distanz : 39 Ereignisse

OW Filter:
<100 km Dist.
<16 mag
>15 ° Höhe
>3km Größe

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	30	100	2	2,5	-20500	2	160	Macro: ctrl+m only OBS	readme!	0,00%		10		S50			2,5	5	deaktivieren	
2	20	A5-B5/2	3		6800	3	320	Macro: ctrl+e ALL	time sort: AM	5,00%		15,5		C11			0,1	3		
3	S50	travel di	4	1000	0,5	C11	640	sort by AM	DAMIT available	0,00%		10		S50			2,5	5		
4	Dist	km S	res n	fram	ms e	fram	T ex	Name Asteroid	Datum Ereignis	Wahrsch	Distar	M komb	M Aste	M Ste	t max	(R) dr	(V) dr	Schat	Sternl	Geme
326	-1	6	9,4	9,0	56	12,5	40	(204850) 2007 RD121	So 02 Nov, 00:12	0,059	2 km S	9,3	20,1	9,3	0,5	10,2	10,8	6 km	51Å˚ SE	4
365	26	16	7,4	11,4	88	25,0	40	(18340) 1989 OM	Mo 03 Nov, 05:16	0	34 km	9,5	18	9,5	1	7,8	8,5	16 km	41Å˚ W	0
596	0	14	9,1	30,2	56	42,5	40	(15138) 2000 EQ93	So 09 Nov, 00:04	0,36	7 km S	8,7	18,6	8,7	1,7	8,6	9,9	14 km	24Å˚ E	2
619	13	6	5,7	7,9	101	20,0	40	(71631) 2000 EX75	Mo 10 Nov, 21:38	0	16 km S	10	17,2	10	0,8	6,5	7,2	6 km	69Å˚ SE	0
731	29	8	12,0	40,0	20	20,0	40	(130239) 2000 CU39	Sa 15 Nov, 20:26	0	33 km	7,5	20,2	7,5	0,8	12,1	12,7	8 km	73Å˚ S	1
1056	-8,5	19	7,7	23,2	69	40,0	40	(35310) 1997 AX1	Do 27 Nov, 19:22	1	1 km N	9	17,6	9	1,6	7,8	8,6	19 km	28Å˚ NE	3
1097	18	8	5,7	38,0	26	25,0	40	(10562) 1993 UB1	Sa 29 Nov, 02:11	0	22 km	8,7	16	8,7	1	6,6	7,3	8 km	49Å˚ NW	0
1279	0	24	7,4	80,0	20	40,0	40	(15410) 1997 YZ	Fr 05 Dez, 02:18	0,559	12 km E	8,2	17	8,2	1,6	8,4	8,8	24 km	50Å˚ SW	1
1683	5	8	8,4	6,5	78	12,5	40	(196091) 2002 TG103	Do 18 Dez, 22:12	0,225	9 km S	9,4	19	9,4	0,5	8,7	9,6	8 km	41Å˚ SE	0
1805	23	18	10,4	65,0	20	32,5	40	(12690) Kochimiraikagak	Mo 22 Dez, 02:05	0	32 km	6,3	16,2	6,3	1,3	9,4	9,9	18 km	47Å˚ S	0
1884	2,5	5	8,4	3,6	111	10,0	40	(43652) 2002 EQ88	Mi 24 Dez, 04:07	0,002	5 km S	9,4	18,6	9,4	0,4	8,3	9,2	5 km	28Å˚ SW	0
1933	14,5	109	3,6	59,6	106	157,5	40	(676) Melitta	Do 25 Dez, 02:10	0	69 km	9,8	14,6	9,8	6,3	4,1	4,8	109 km	50Å˚ S	18
2076	25	6	7,8	2,6	114	7,5	40	(22085) 2000 AA169	Mo 29 Dez, 22:29	0	28 km	9,5	18,2	9,5	0,3	8,1	8,7	6 km	31Å˚ SE	1
2319	16	8	9,0	2,9	170	12,5	40	(83282) 2001 RD88	So 04 Jan, 03:17	0	20 km	9,9	19,7	9,9	0,5	8,8	9,8	8 km	33Å˚ W	1
2548	0,5	5	8,4	4,2	95	10,0	40	(106810) 2000 XY39	Sa 10 Jan, 17:33	0,437	3 km S	9,2	18,4	9,2	0,4	8,4	9,2	5 km	26Å˚ NE	1
2679	4	6	6,9	35,8	22	20,0	40	(71124) 1999 XR173	Di 13 Jan, 21:52	0,191	7 km N	8,9	17,8	8,9	0,8	7,9	8,9	6 km	72Å˚ S	2
2886	-1,5	5	8,8	5,4	73	10,0	40	(117782) 2005 GL115	Di 20 Jan, 21:57	0,274	1 km E	9,7	20	9,7	0,4	9,5	10,3	5 km	65Å˚ SE	1
3164	27	28	6,9	16,8	155	65,0	40	(3372) Bratijchuk	Fr 30 Jan, 01:08	0	41 km S	9,4	16,7	9,4	2,6	6,6	7,3	28 km	20Å˚ W	0
3284	13	12	11,6	35,0	20	17,5	40	(163407) 2002 RH5	Di 03 Feb, 22:45	0,14	19 km E	7,2	19,2	7,2	0,7	11,4	12	12 km	30Å˚ SE	0
3307	21,5	5	7,8	9,6	52	12,5	40	(88972) 2001 TT58	Mi 04 Feb, 23:15	0	24 km	9,4	18,8	9,4	0,5	8,6	9,4	5 km	62Å˚ SE	0
3364	-2,5	9	10,8	22,5	27	15,0	40	(20710) 1999 XP10	Sa 07 Feb, 00:45	0,997	2 km N	7,8	19,3	7,8	0,6	10,5	11,5	9 km	17Å˚ W	0
3483	27,5	5	8,9	6,6	76	12,5	40	(104924) 2000 JL21	Mo 16 Feb, 21:33	0	30 km	9,9	20,5	9,9	0,5	9,9	10,6	5 km	81Å˚ SE	1
3728	2,5	41	10,5	100,0	20	50,0	40	(5368) Vitagliano	Do 12 Mrz, 21:06	0,066	23 km	7	17,7	7	2	9,8	10,7	41 km	39Å˚ W	2

OW Cloud Filter



JPL Seite

5368 Vitagliano (1984 SW5)

Classification: Outer Main-belt Asteroid **SPKID:** 20005368 **Related Links:** Ephemeris

Orbit Viewer [show]

Orbit Parameters [show]

Physical Parameters [hide]

Parameter	Value	Units	Sigma	Reference	N
[H] absolute magnitude	11.13			MPO914192	
diameter	34.807	km	0.415	urn:nasa:pds:neowise_diam...	
rotation period	59.36	h		LCDB (Rev. 2023-October);...	[Result based on less than full coverage...
geometric albedo	0.058		0.011	urn:nasa:pds:neowise_diam...	

[Result based on less than full coverage, so that the period may be wrong by 30 percent or so.] REFERENCE LIST:[Warner, B.D.; Stephens, R.D.; Coley, D.R. (2017) Minor Planet Bull. 44, 130-137.]

https://ssd.jpl.nasa.gov/horizons

(5368) Vitagliano – DAMIT (Uni Prag)

Database of Asteroid Models from Inversion Techniques

damit.cuni.cz/projects/damit/?q=5368

DAMIT Models Tumblers Tables Export Documentation More

Modell 13151 vs. 13152:

Unterschied am 12.03.2026: 0,33 Umdrehungen (118,6°)

P: ± 0,001h entspricht ±0,11Umdrehungen(±39,5°)

5368

(10757 asteroid

Models

	λ	β	P	Files
(5368) Vitagliano				LCref | lc.txt | lc.jsc
View model	35	62	30.281	IAUspin | shape.pr
View model	205	63	30.278	IAUspin | shape.pr

vgl. JPL: 59,36h, DAMIT 29,69h: Faktor 2 mit Δ -2%

https://damit.cuni.cz **Ďurech** & Hanuš (2023)

damit.cuni.cz/projects/damit/asteroid_models/view/13151

DAMIT Models Tumblers Tables Export Documentation More

(5368) Vitagliano – Model 13151

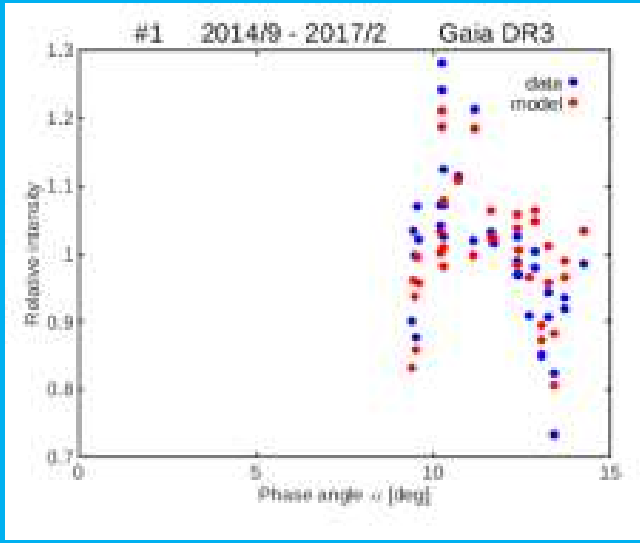
Details

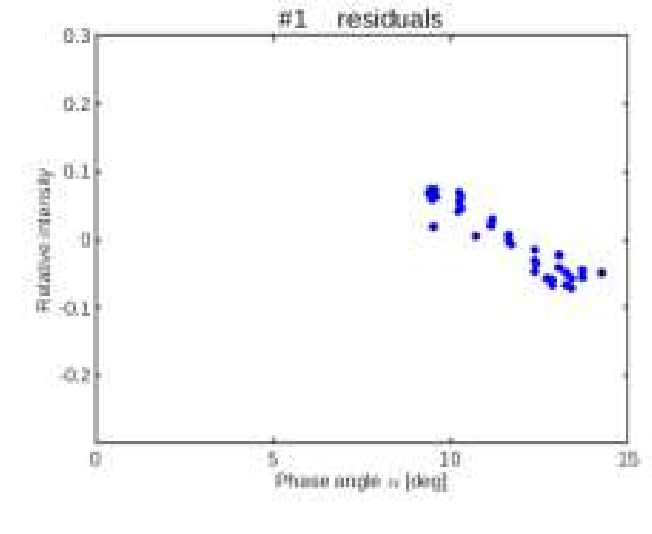
ID	13151
Asteroid	(5368) Vitagliano
λ	35
β	62
P	30.281
YORP	
t_0	2456916
φ_0	0
Light Scattering Model	LSL
p_1	0.1
p_2	0.5
p_3	0.1
p_4	-0.68

15.09.2014

(5368) Vitagliano – ISAM

Infrared Standard Asteroid Model






```
Satellite IR diameters for asteroid (5368)

  NEOWISE:  34.8 ± 3.2km:  albedo 0.058
     AcuA:  39.3 ± 7.6km:  albedo 0.046

Weighted diameter:  35.5 ±  3.2 km
[including recommended increase in uncertainty]

Individual diameter and albedo measurements
   34.8 ±  3.2km:  albedo 0.058:  NEOWISE
   39.3 ±  8.4km:  albedo 0.046:  AcuA

Estimated diameter, absolute mag. H0 = 11.17
    albedo 0.40 => diameter = 12 km
    albedo 0.35 => diameter = 13 km
    albedo 0.30 => diameter = 14 km
    albedo 0.25 => diameter = 16 km
    albedo 0.20 => diameter = 17 km
  * albedo 0.15 => diameter = 20 km *
    albedo 0.10 => diameter = 25 km
    albedo 0.09 => diameter = 26 km
    albedo 0.08 => diameter = 27 km
    albedo 0.07 => diameter = 29 km
    albedo 0.06 => diameter = 32 km
    albedo 0.05 => diameter = 35 km
    albedo 0.04 => diameter = 39 km
    albedo 0.03 => diameter = 45 km
    albedo 0.02 => diameter = 55 km
    albedo 0.01 => diameter = 78 km
  Trojans 0.06 => diameter = 32 km
```

Light curve fits für (5368) model 13152

Event JD = 2461112.3375

Tägliche Rotationsrate= 285.355038°/Tag (= model 13152)

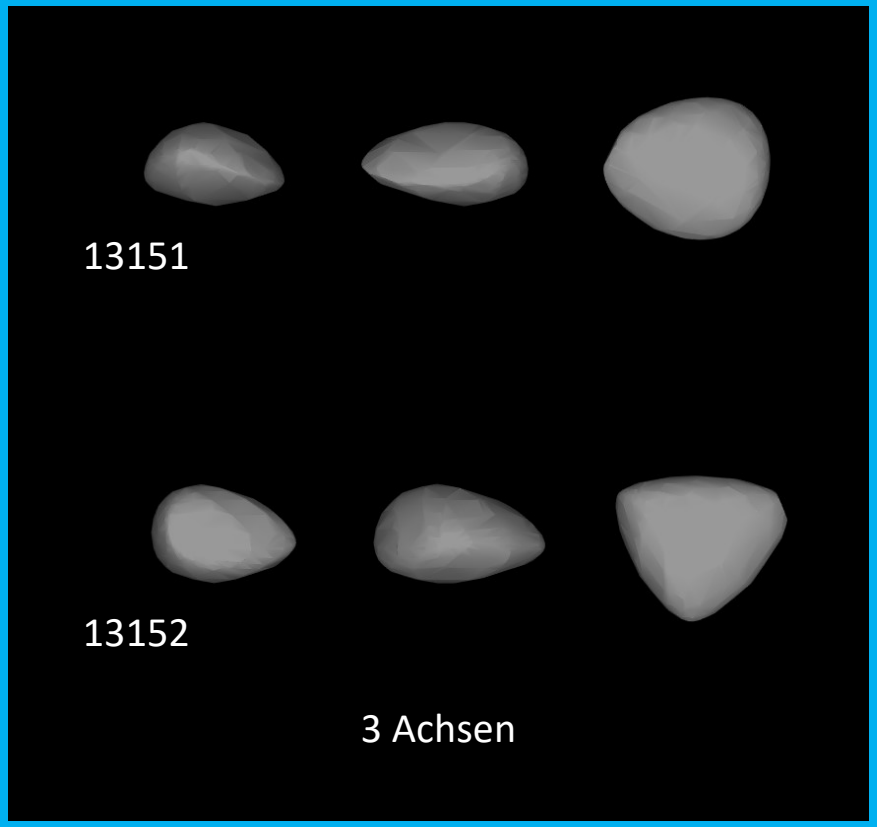
Anm : 0,000001°/Tag entspricht ± 1,3 Sek.

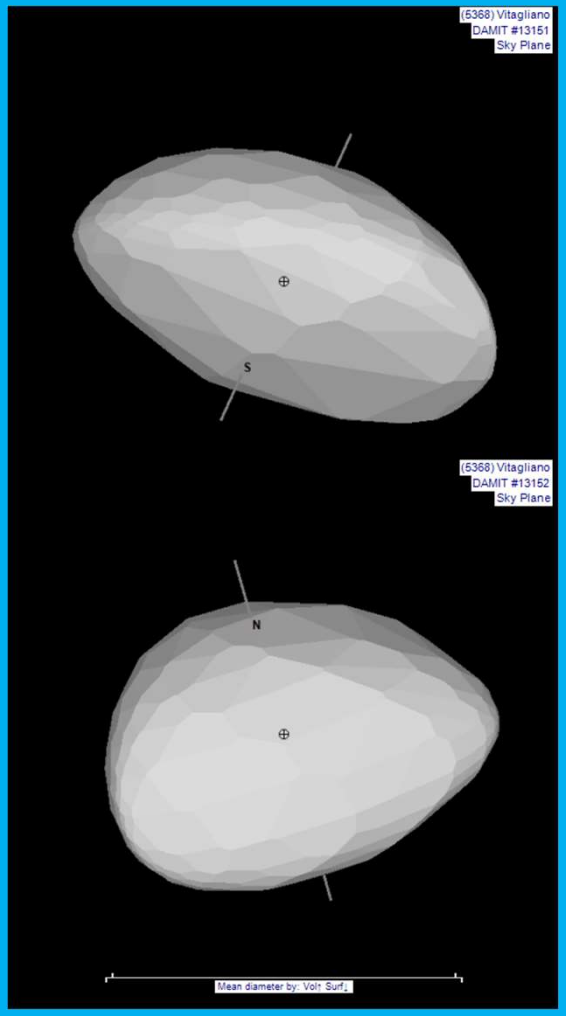
Occult v4.2025.9.10

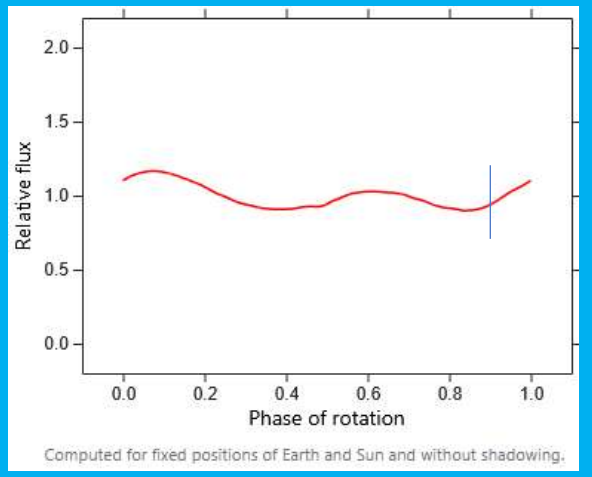
NEOWISE: 2009-2013/2024 (genauer)

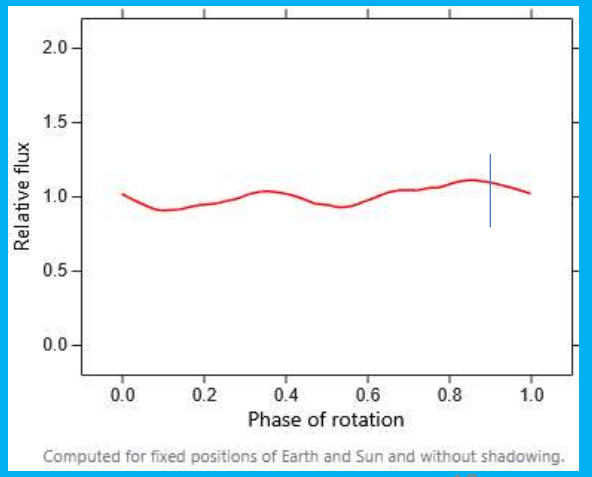
AcuA = Asteroid Catalog using AKARI: 2006 - 2011 (Ergänzung)

(5368) Vitagliano – DAMIT Modell 13151, 13152 (occult4)



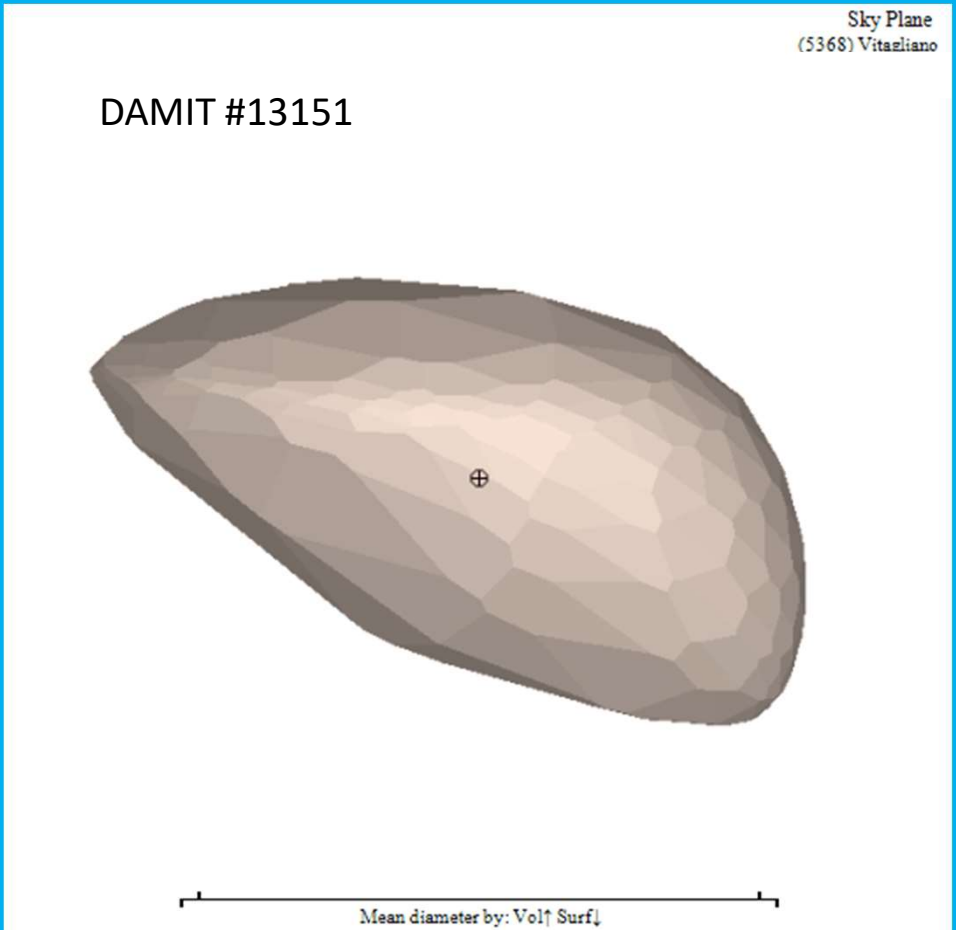




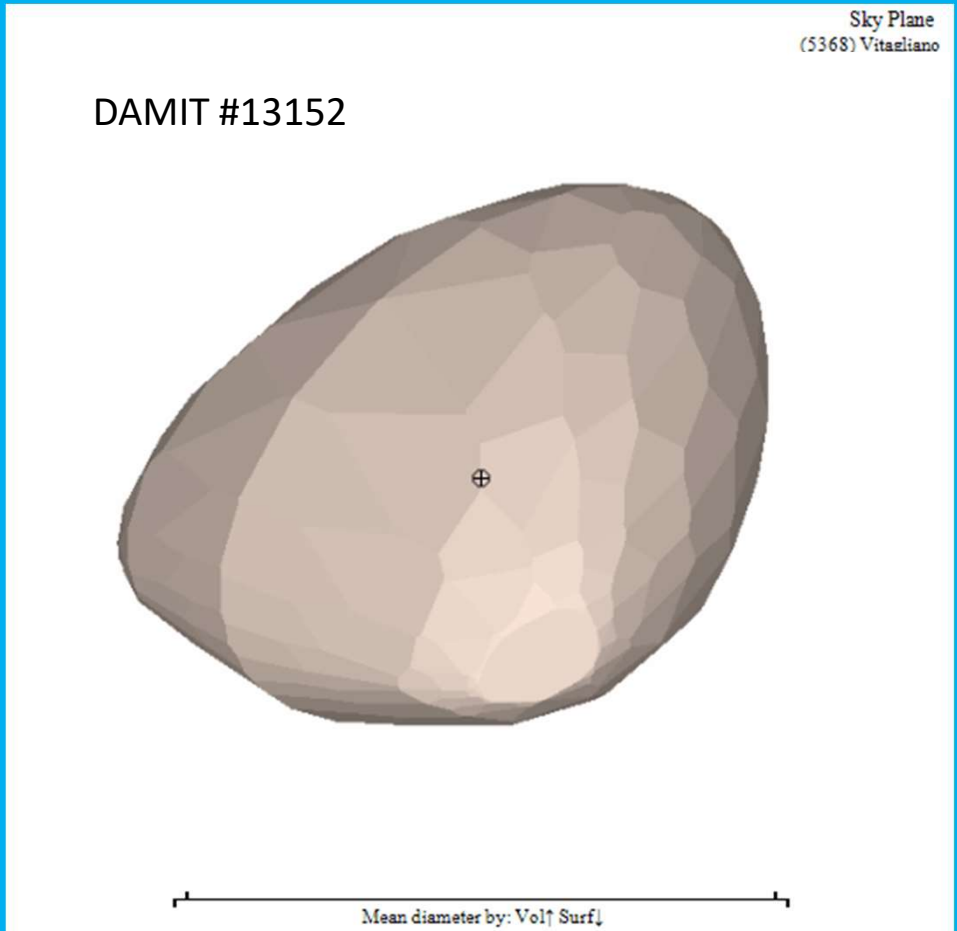


(5368) Vitagliano – animiert 30h

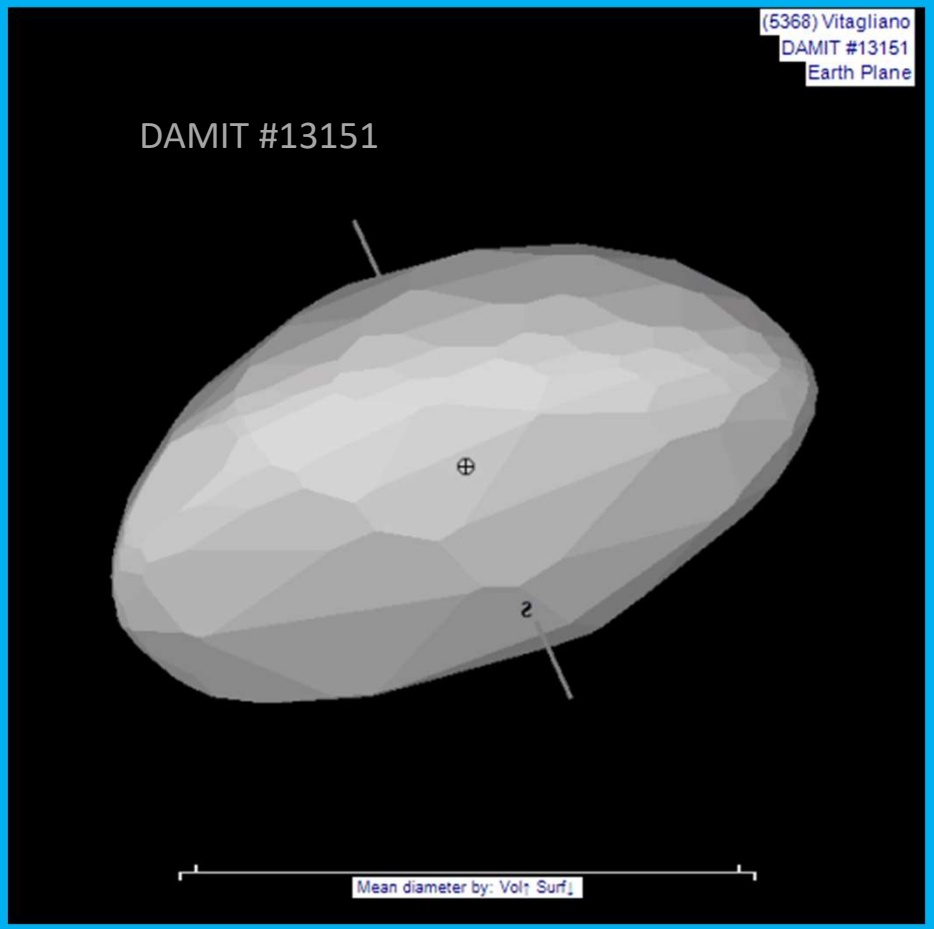
DAMIT #13151

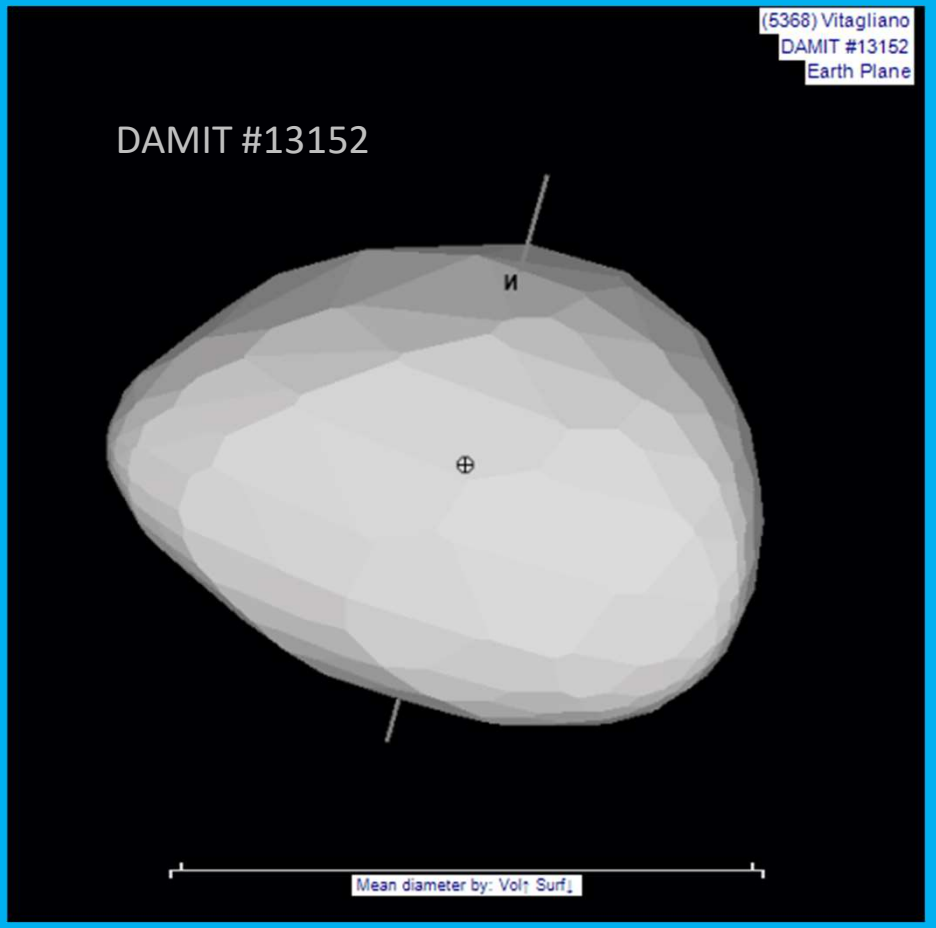


DAMIT #13152

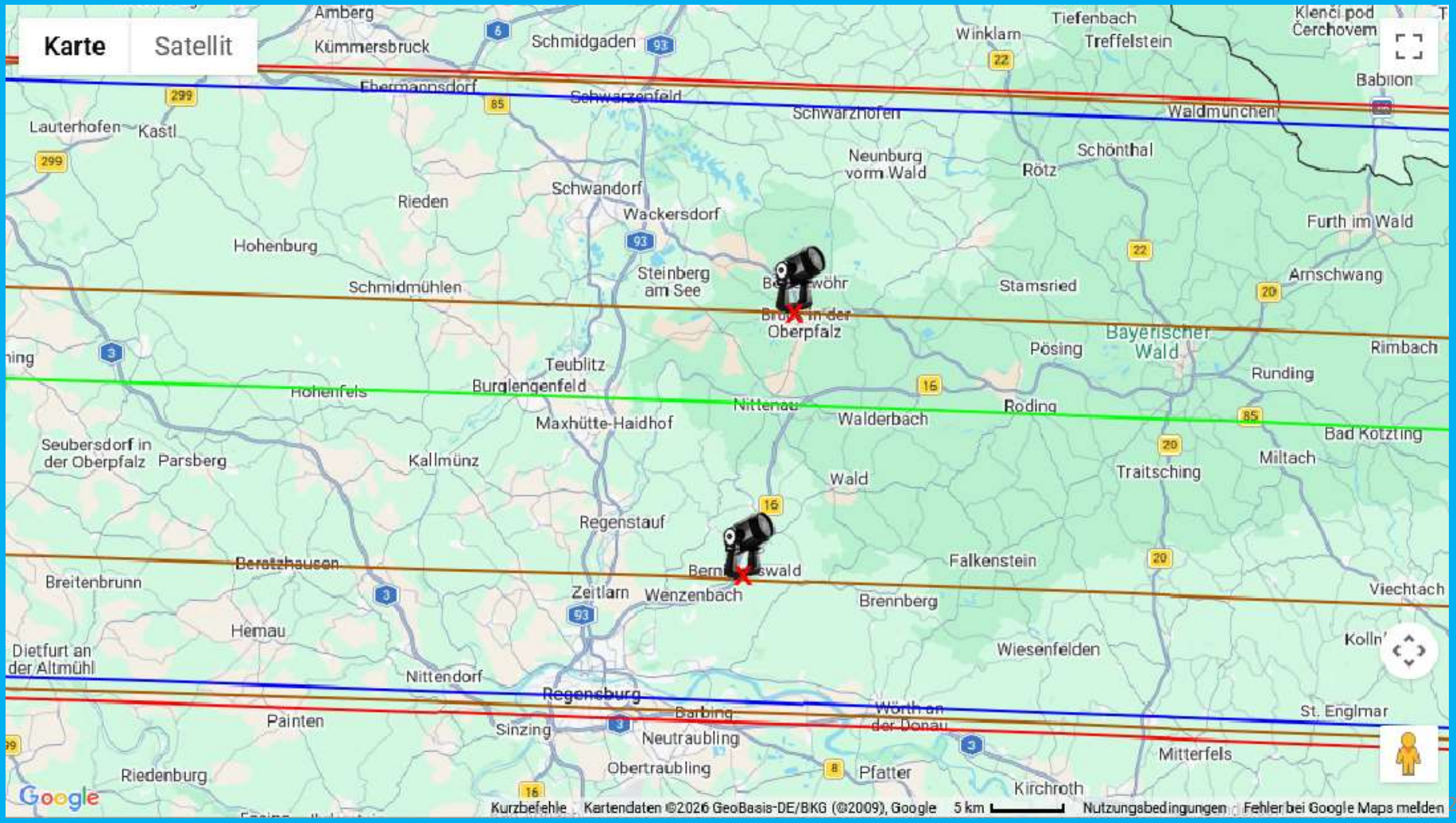


(5368) Vitagliano – animiert ±5° / ±25 Min.

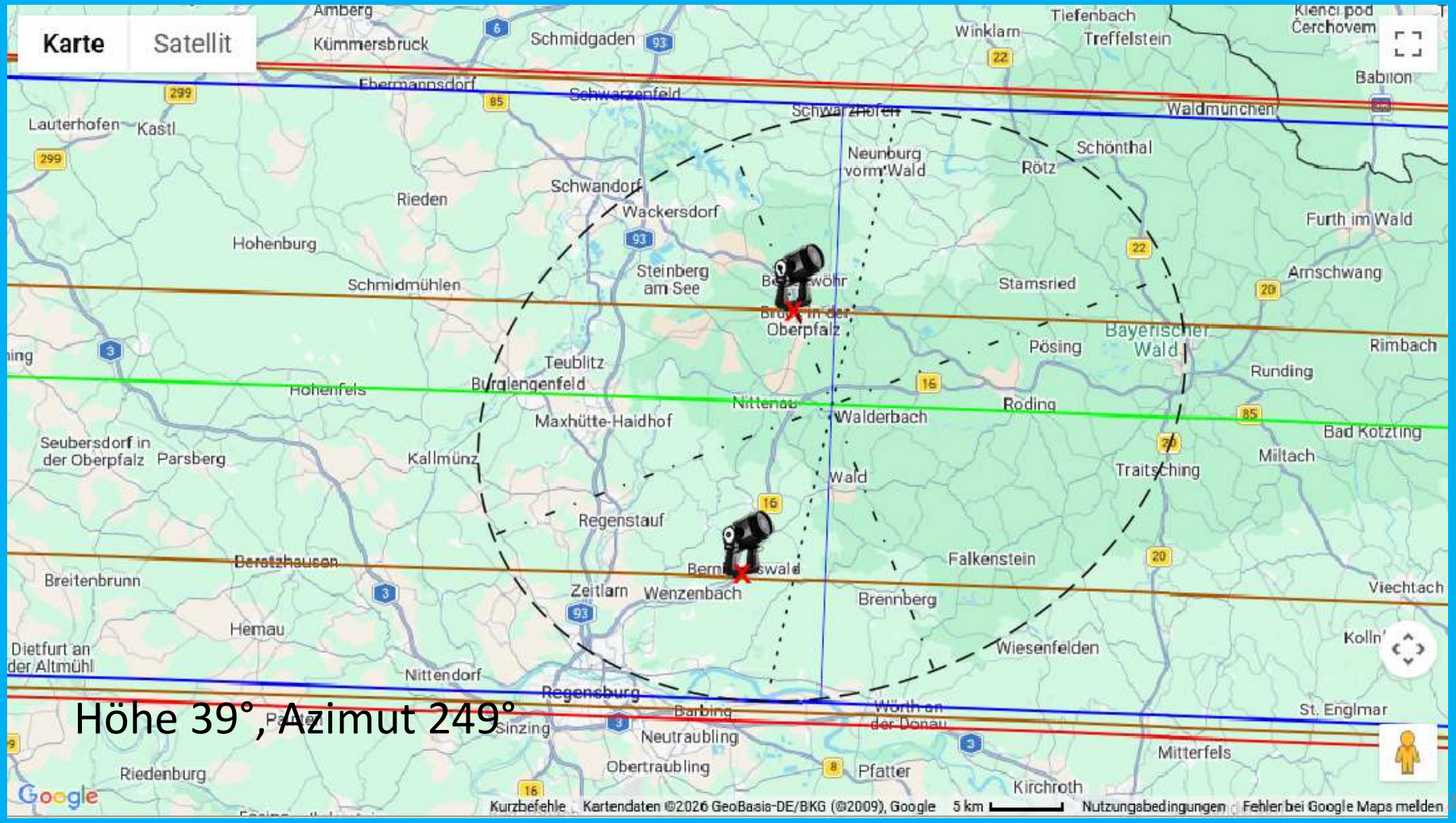




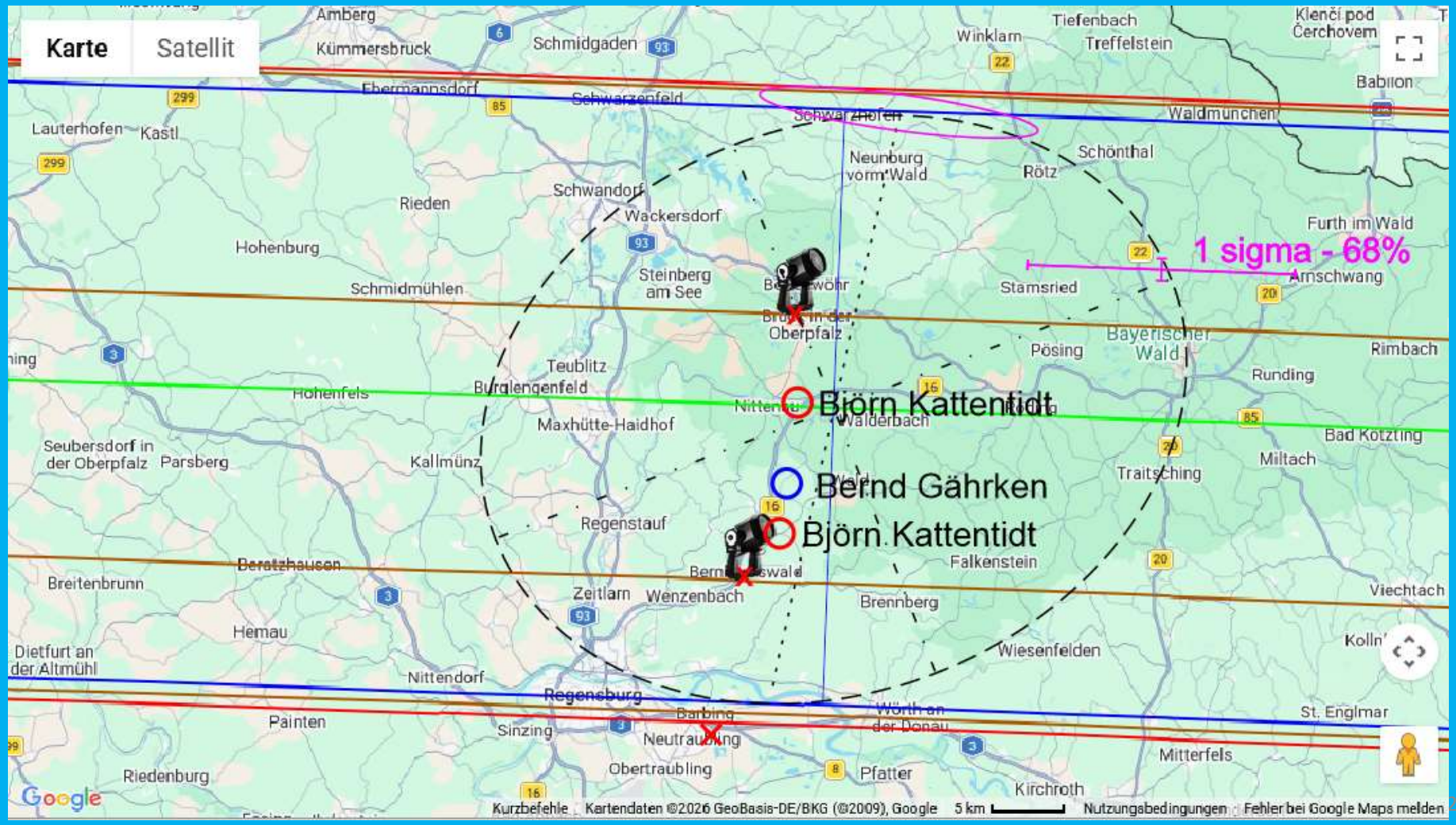
Expedition: Bedeckungspad und Standortplanung



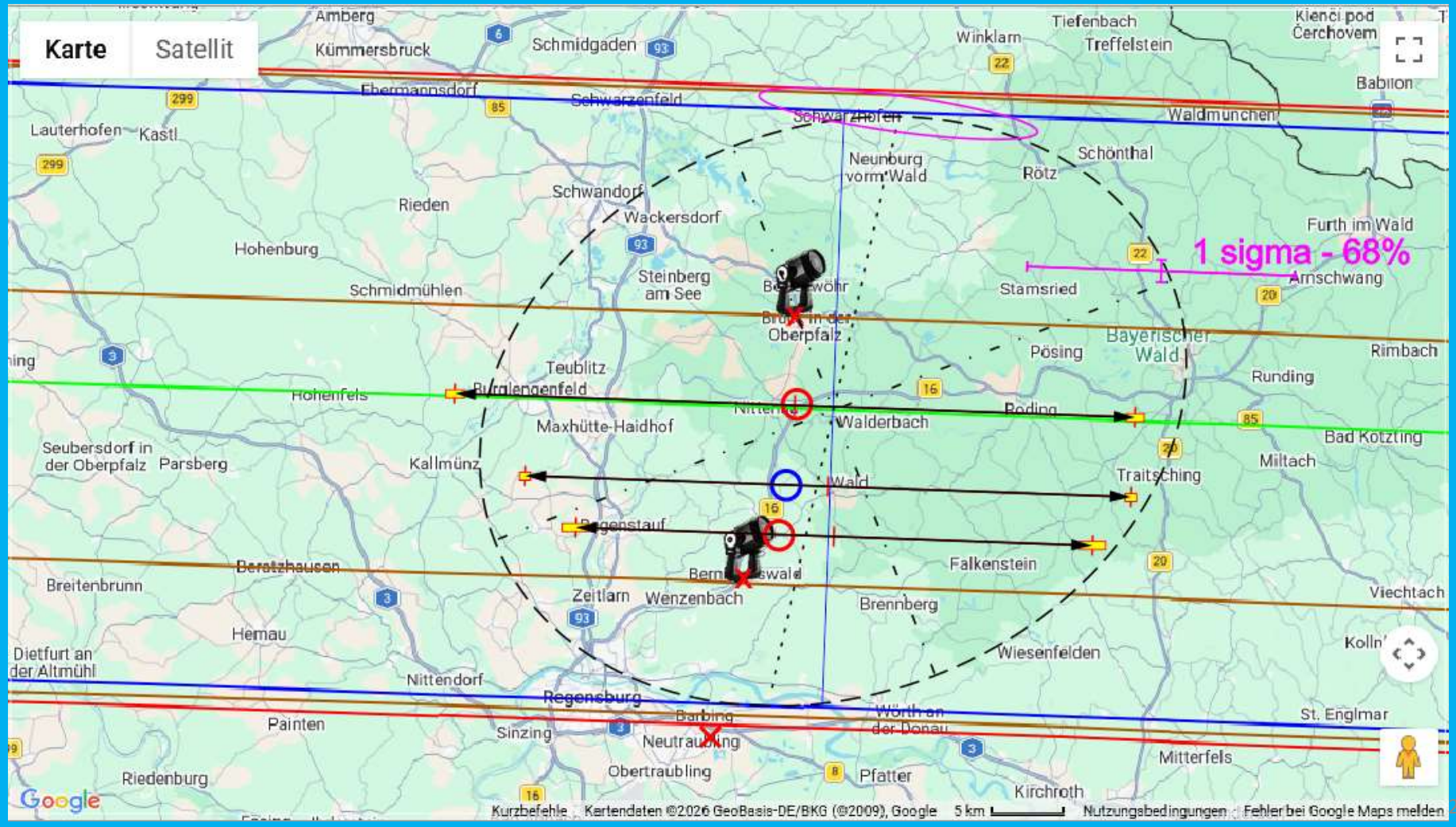
Projektion als kugelförmiger Asteroid



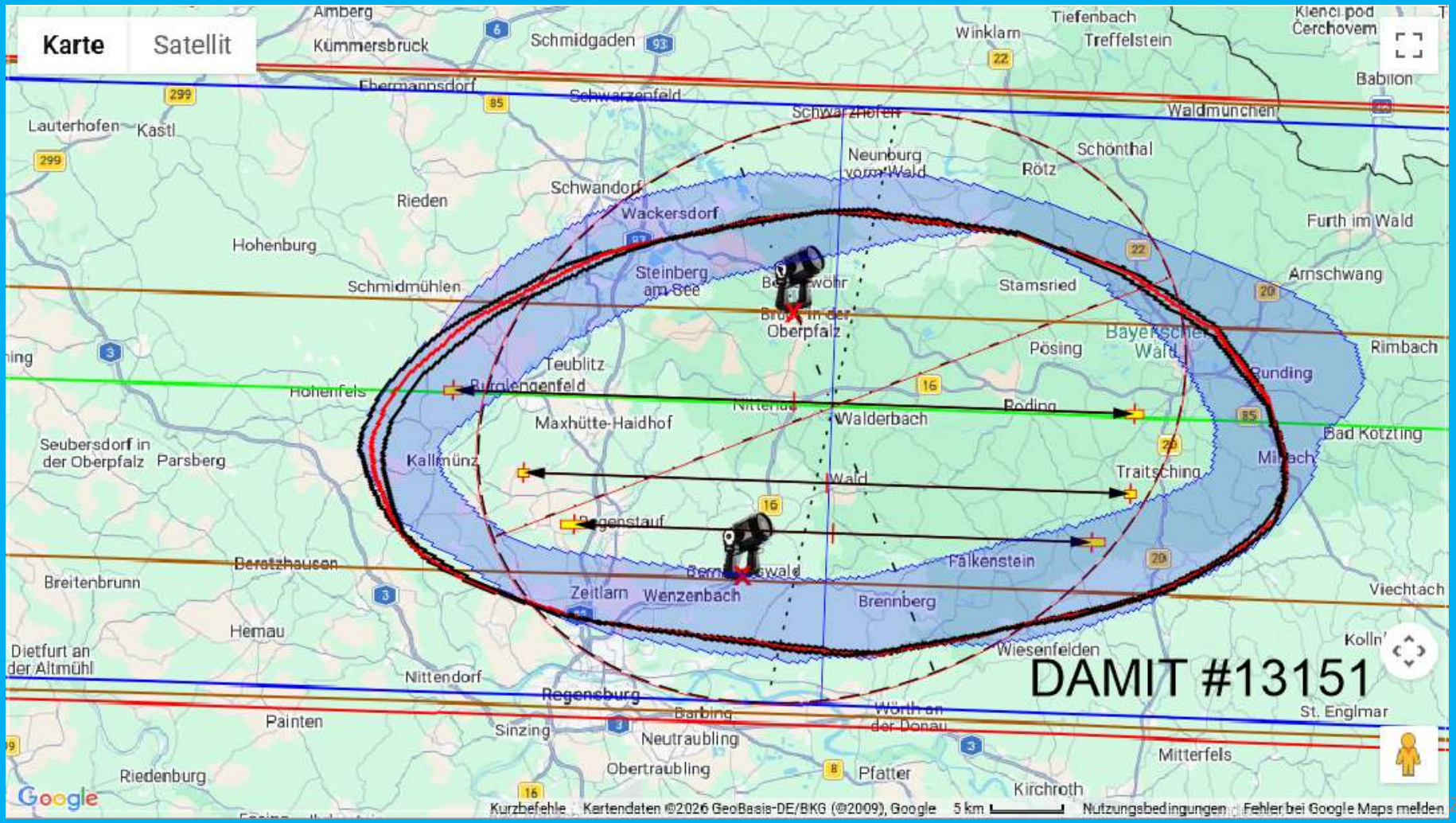
Tatsächliche Standorte



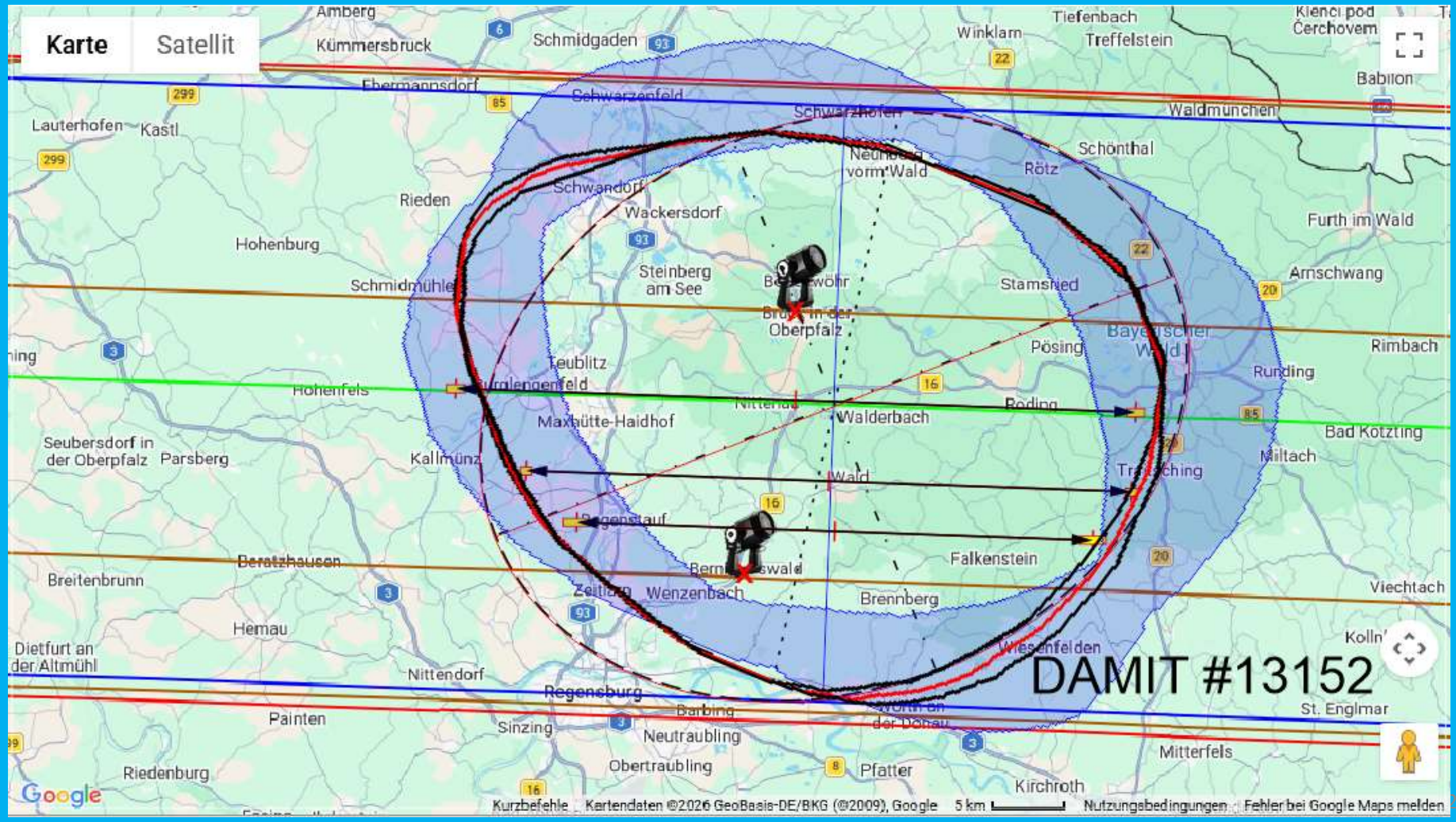
Gemessene Zeiten korrigiert projiziert



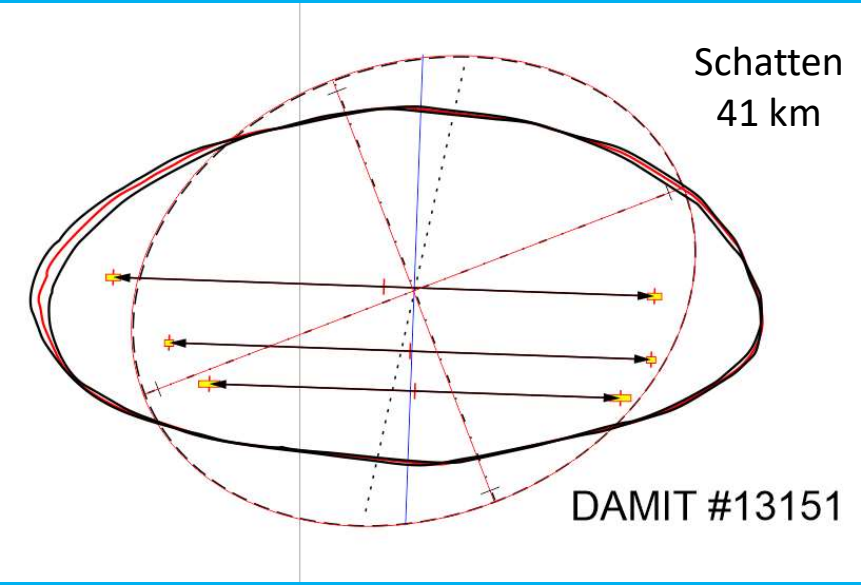
(5368) Vitagliano – Modell #1: Vorhersage - Beobachtung

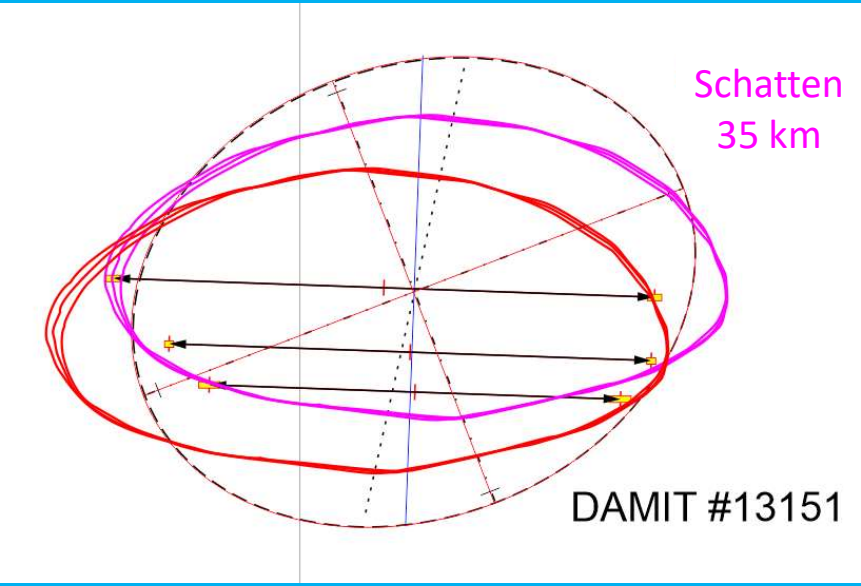


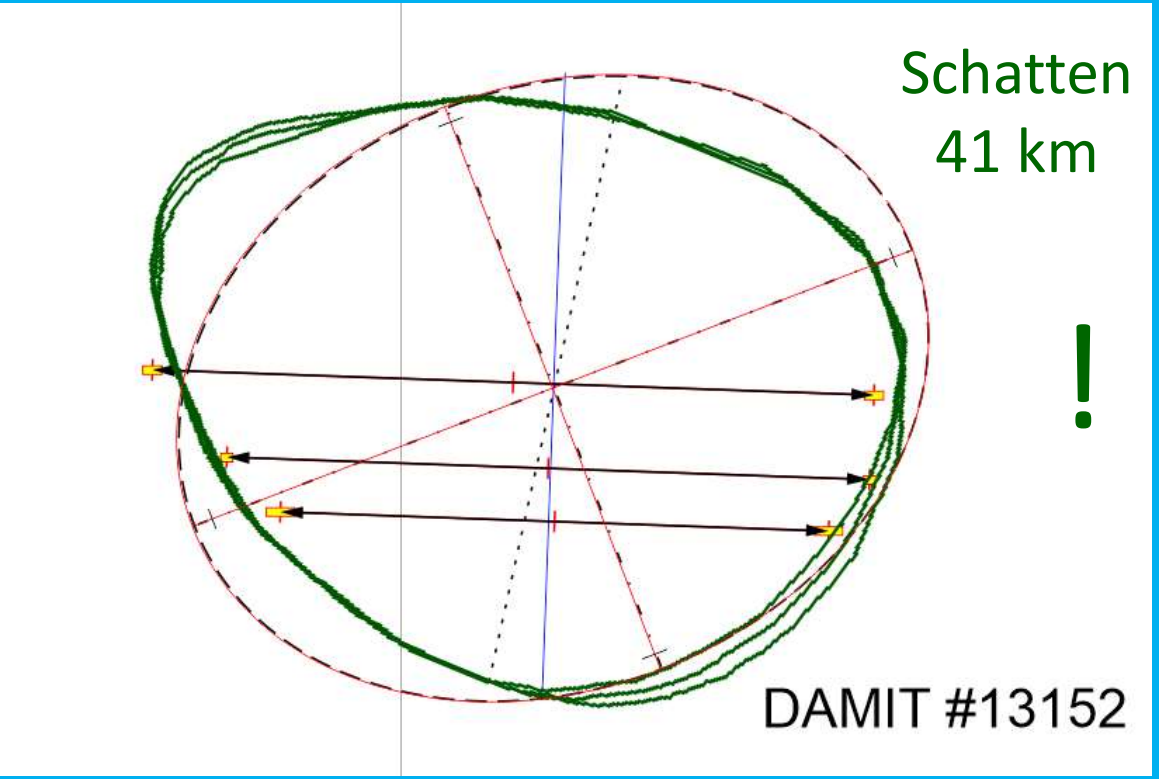
(5368) Vitagliano – Modell #2 : Vorhersage - Beobachtung



Ergebnis?


Schatten
41 km
DAMIT #13151


Schatten
35 km
DAMIT #13151


Schatten
41 km
!
DAMIT #13152

(5368) Vitagliano – Modell #2:
bisherige Beobachtungen – eigenes Ergebnis ungeprüft

```
Observed chord lengths

Length       # Observer

5368 Vitagliano       2020 Apr  2       UCAC4 447-054339
  13.9 km    1 D Gault

5368 Vitagliano       2022 Oct  9       UCAC4 360-195775
  37.9 km    1 A Howell

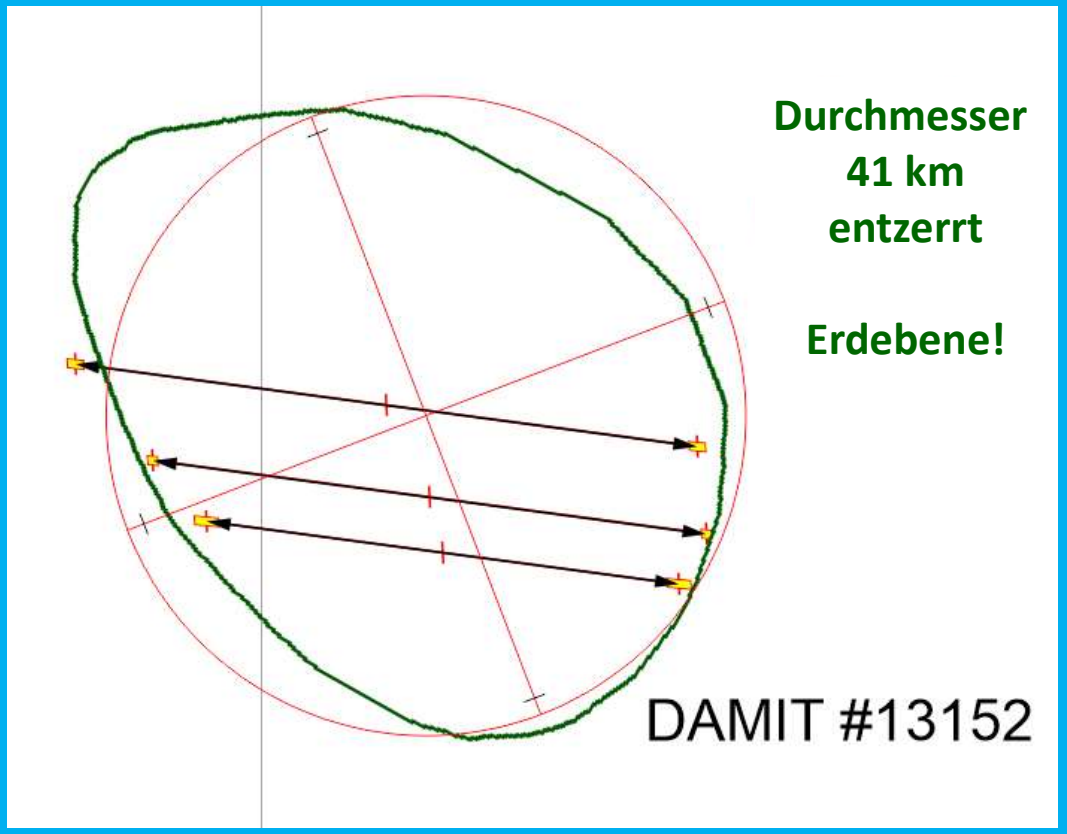
5368 Vitagliano       2023 Sep 21       UCAC4 401-130857
  33.8 km    1 P Maley

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Satellite IR diameters for asteroid (5368)

  NEOWISE:  34.8 ± 3.2km:  albedo 0.058
     AcuA:  39.3 ± 7.6km:  albedo 0.046

Weighted diameter:  35.5 ±  3.2 km
[including recommended increase in uncertainty]
________________________________________
Individual diameter and albedo measurements
   34.8 ±  3.2km:  albedo 0.058:  NEOWISE
   39.3 ±  8.4km:  albedo 0.046:  AcuA
```

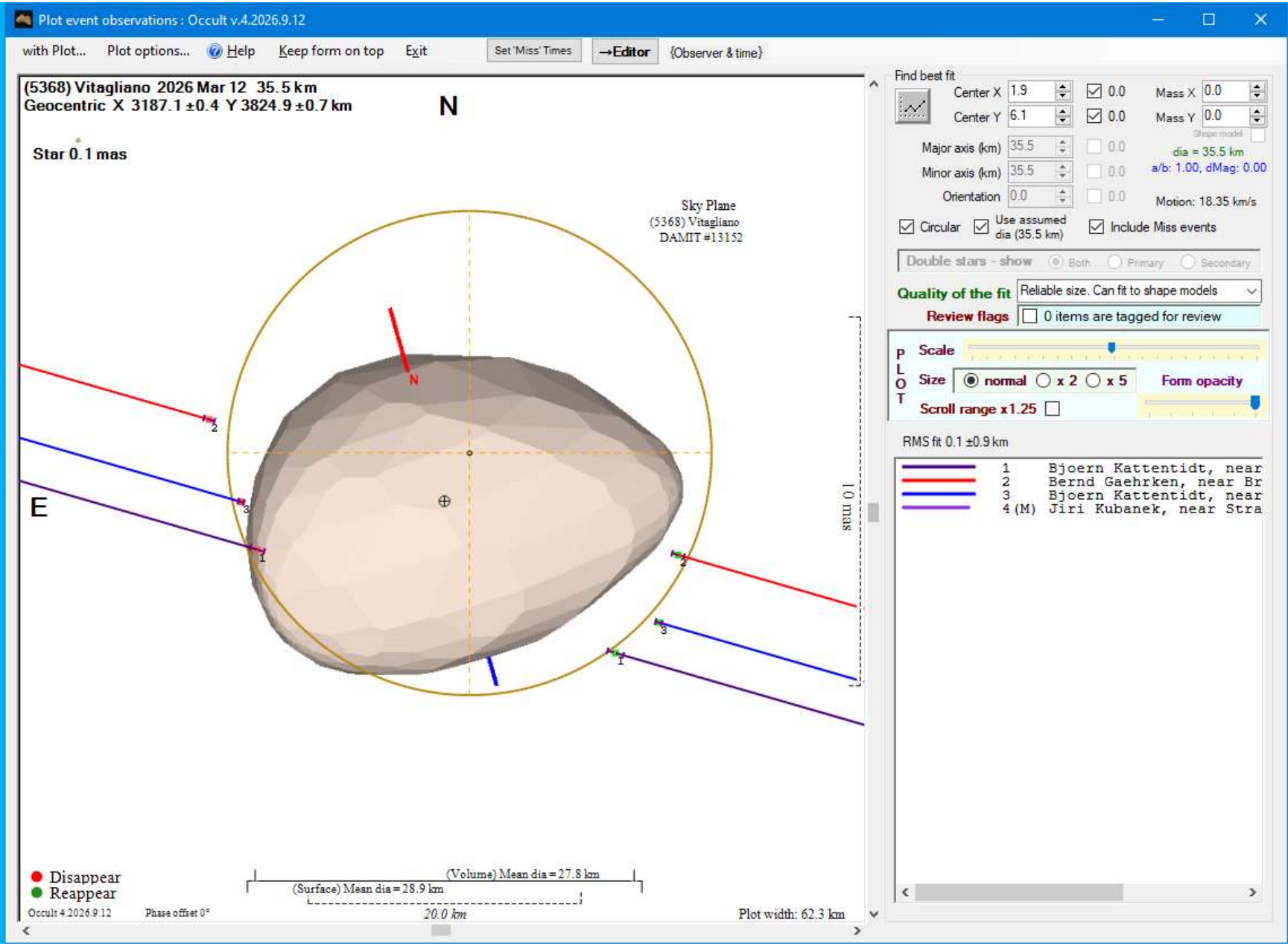


Occult v4.2025.9.10

Occult 4

DAMIT #13152

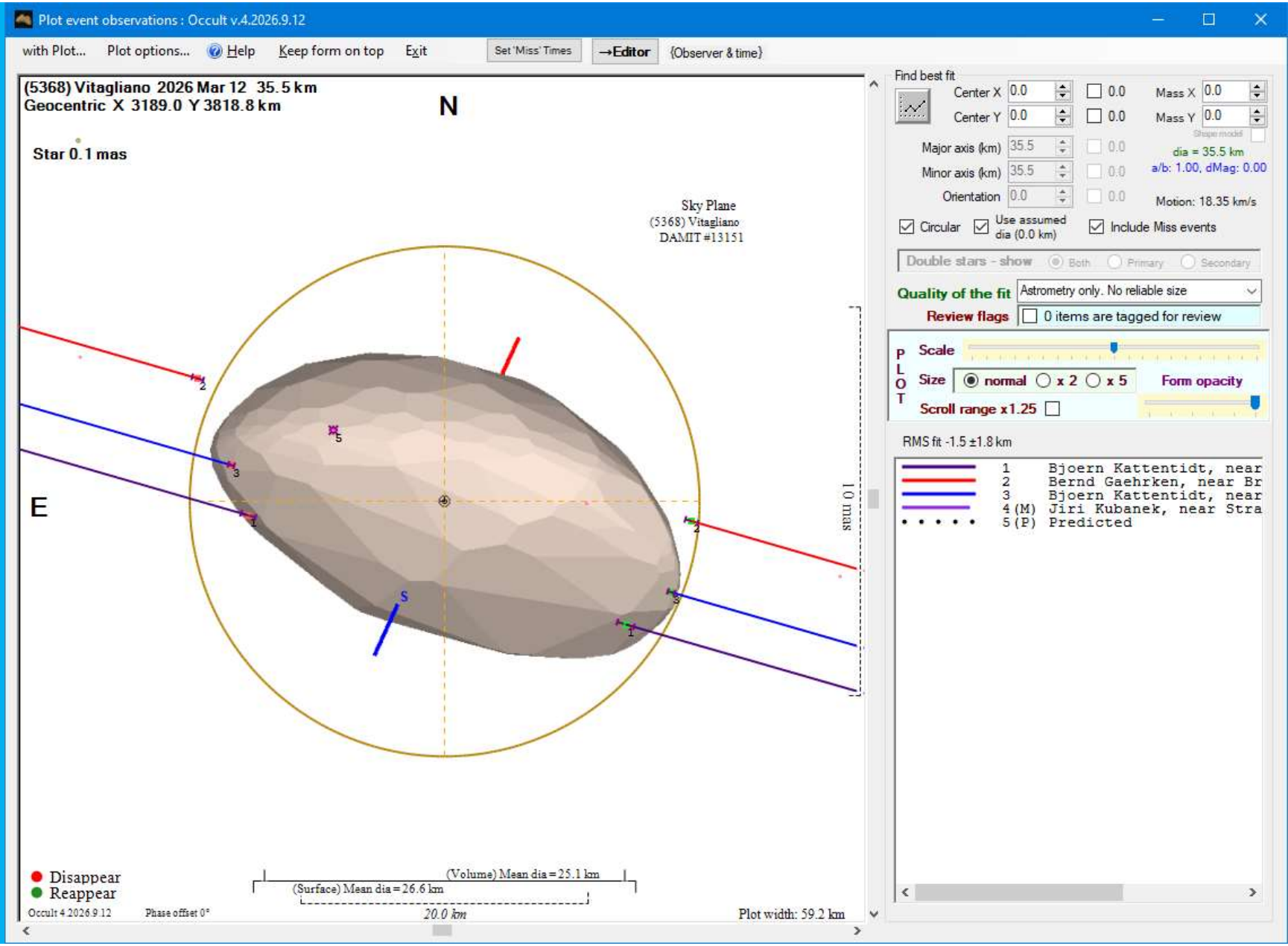
Modell #2



Occult 4

DAMIT #13151

Modell #1

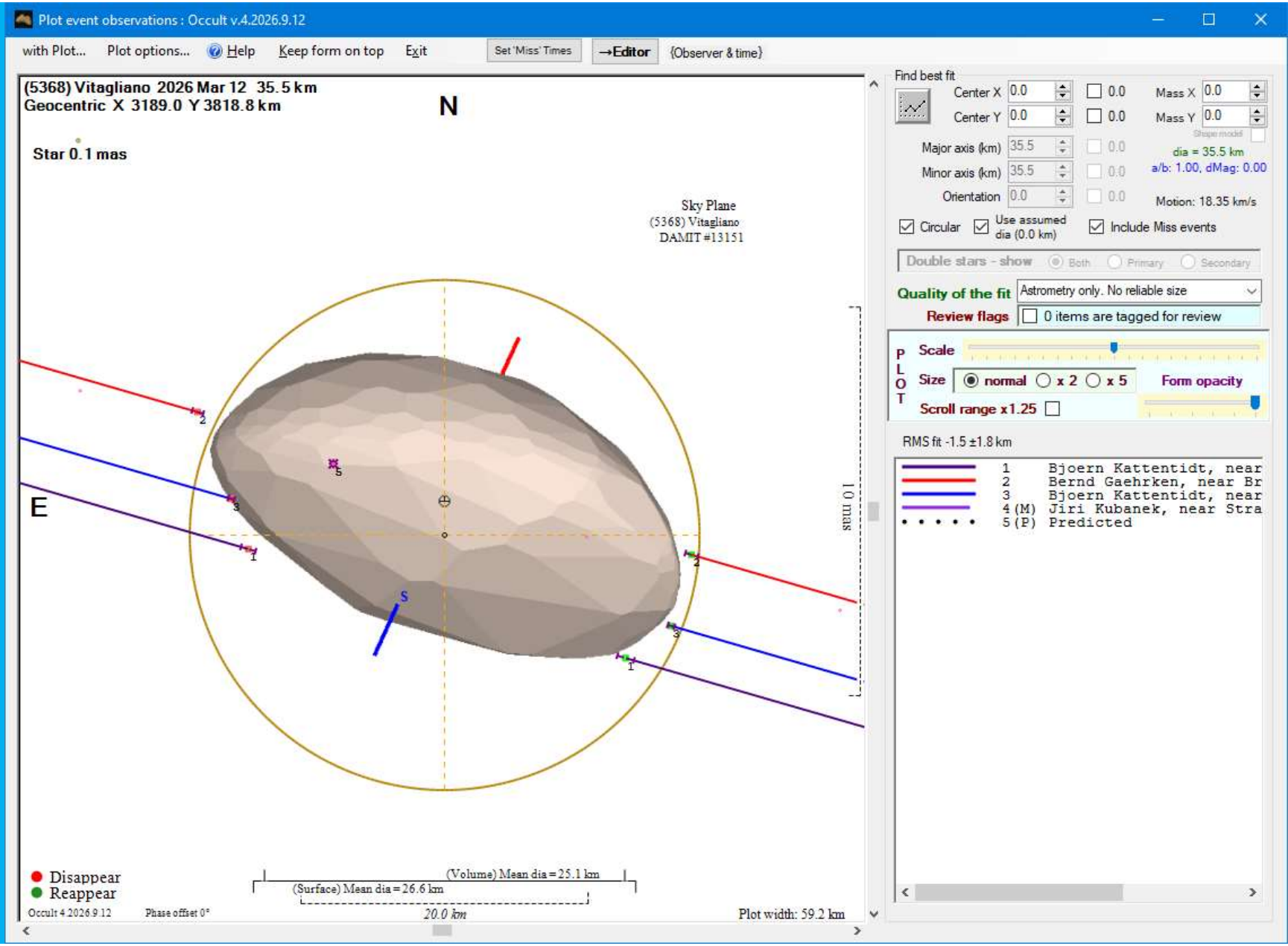


Occult 4

DAMIT #13151

Modell #1

Umriß verschoben

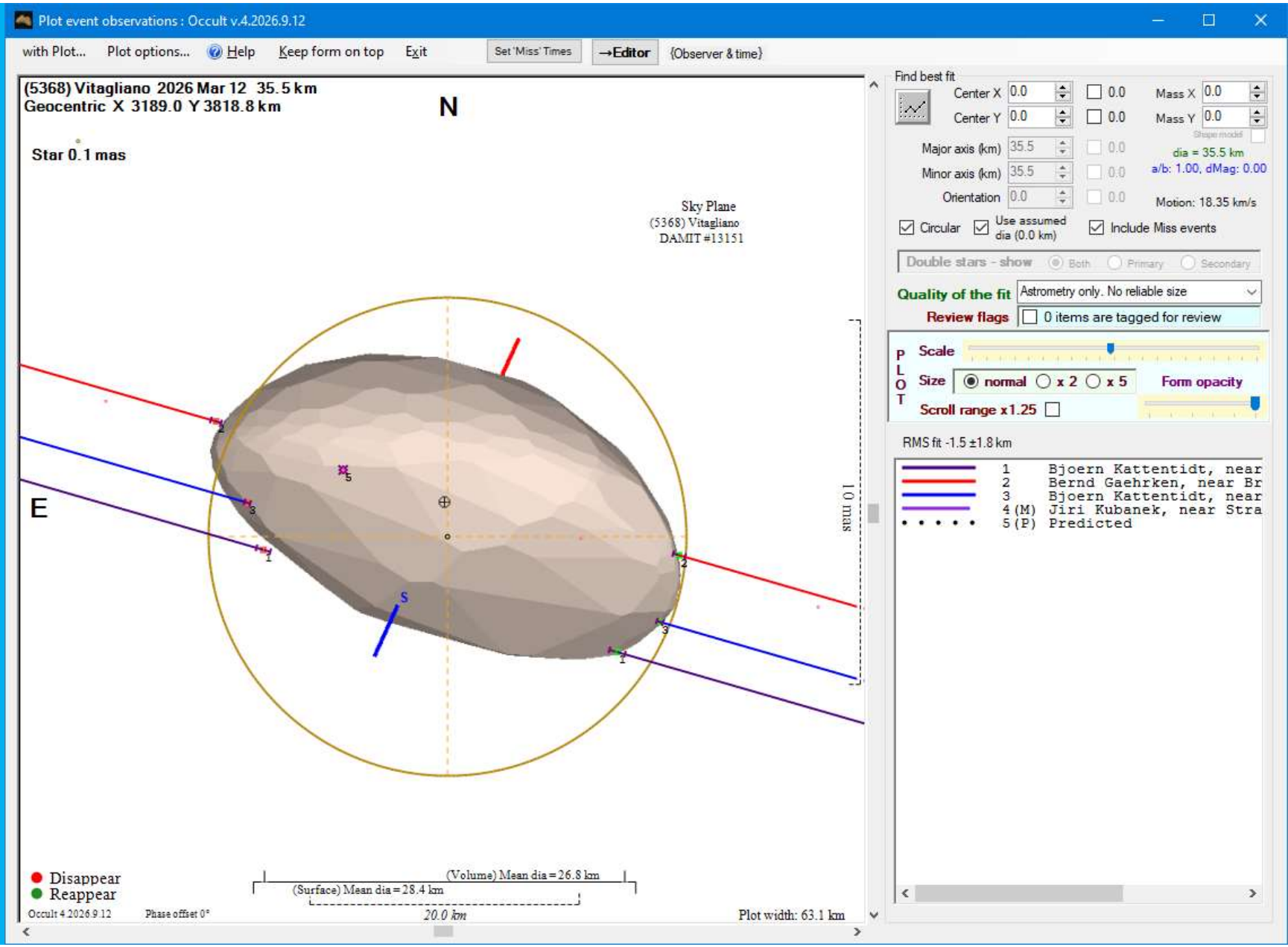


Occult 4

DAMIT #13151

Modell #1

Ellipse verkleinert gemäß Sehnen

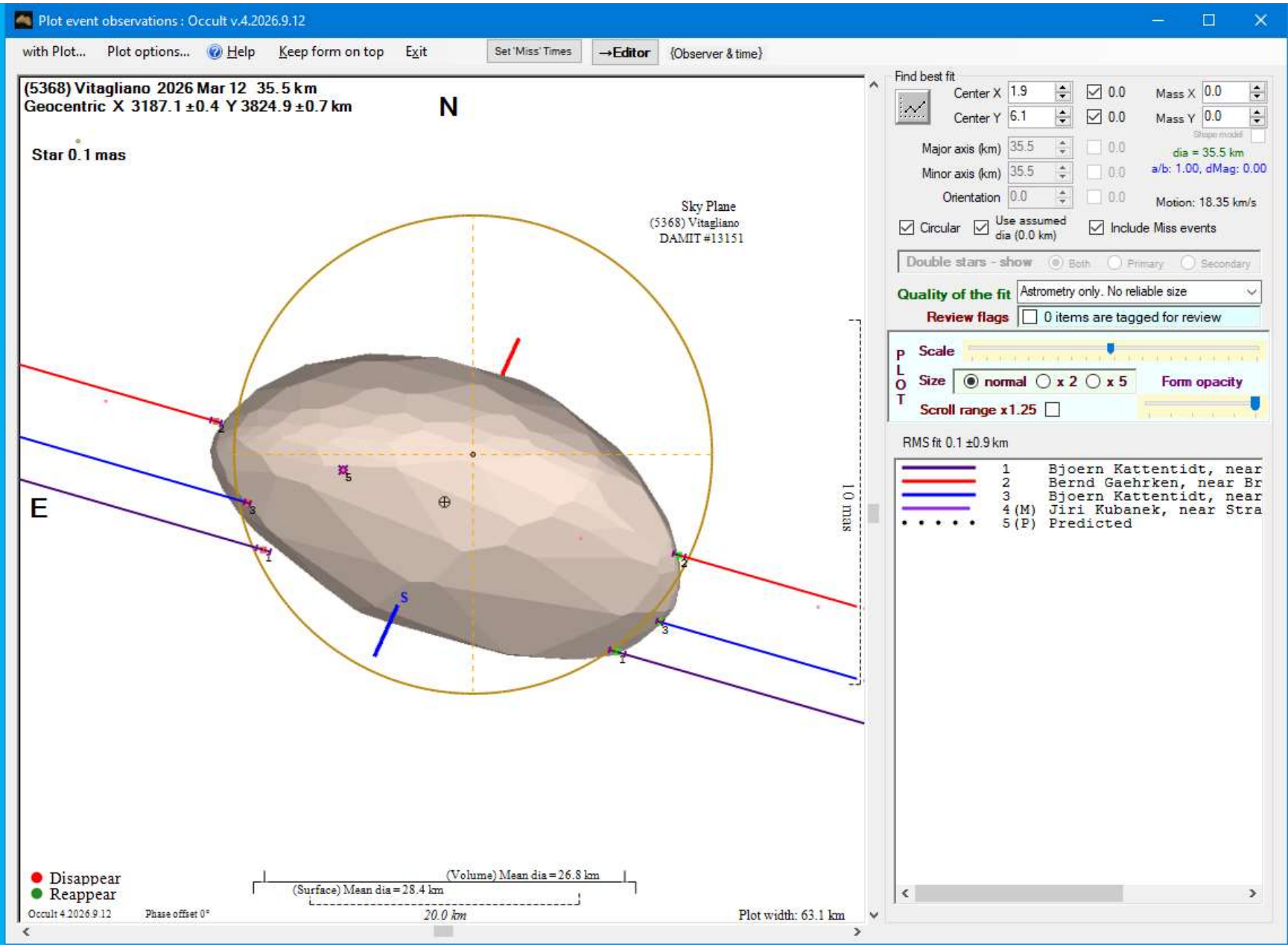


Occult 4

DAMIT #13151

Model #1

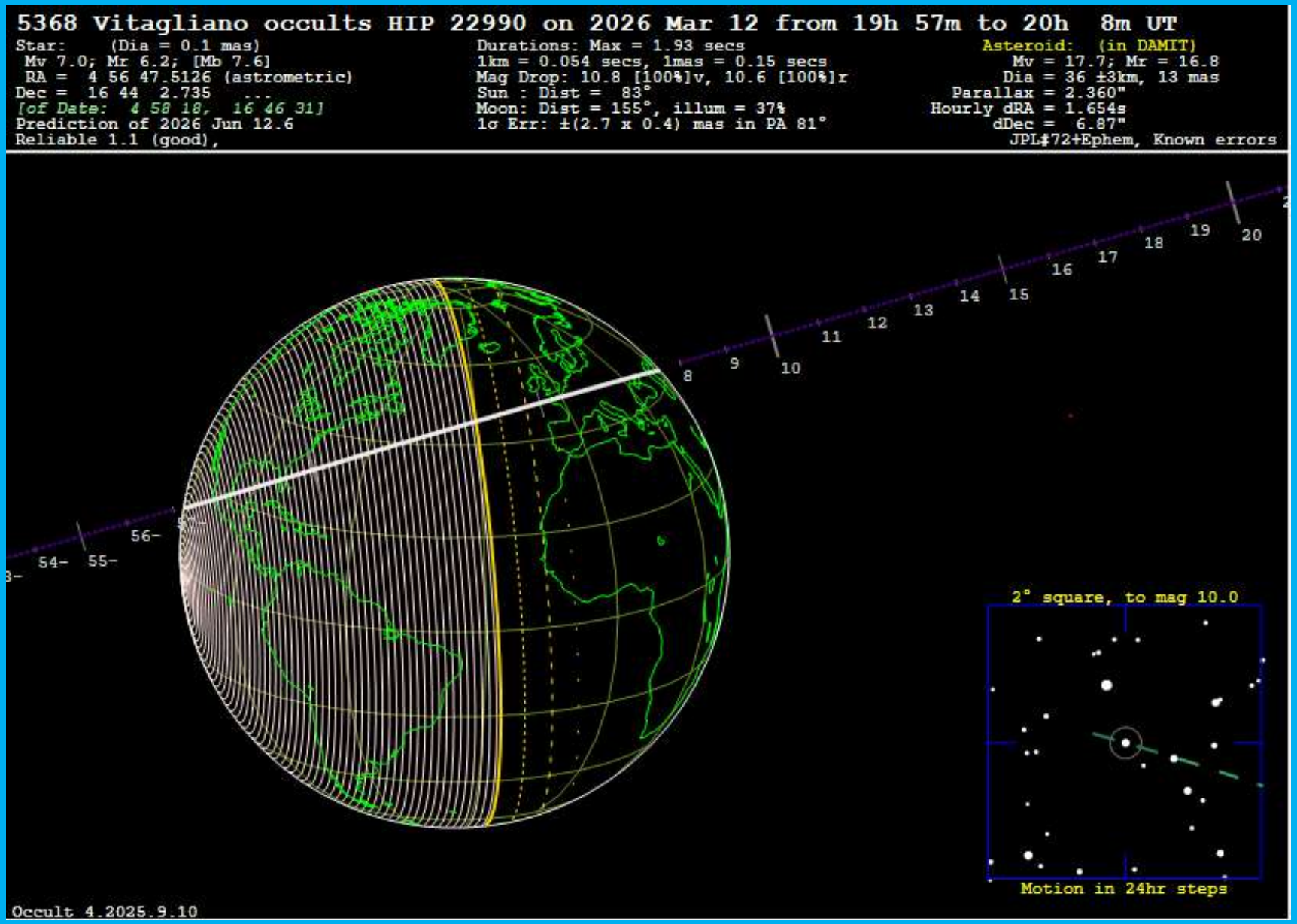
beste Anpassung Ellipse



(5368) Vitagliano – „Modelle“
Maßstab 1 : 1.000.000






5368 Vitagliano occults HIP 22990 on 2026 Mar 12 from 19h 57m to 20h 8m UT
Star: (Dia = 0.1 mas)
Mv 7.0; Mr 6.2; [Mb 7.6]
RA = 4 56 47.5126 (astrometric)
Dec = 16 44 2.735 ...
[of Date: 4 58 18, 16 46 31]
Prediction of 2026 Jun 12.6
Reliable 1.1 (good),
Durations: Max = 1.93 secs
1km = 0.054 secs, 1mas = 0.15 secs
Mag Drop: 10.8 [100%]v, 10.6 [100%]r
Sun : Dist = 83°
Moon: Dist = 155°, illum = 37%
1σ Err: ±(2.7 x 0.4) mas in PA 81°
Asteroid: (in DAMIT)
Mv = 17.7; Mr = 16.8
Dia = 36 ±3km, 13 mas
Parallax = 2.360"
Hourly dRA = 1.654s
dDec = 6.87"
JPL#72+Ephem, Known errors
8
9
10
11
12
13
14
15
16
17
18
19
20
54-
55-
56-
2° square, to mag 10.0
Motion in 24hr steps
Occult 4.2025.9.10